SUBMODULAR MAXIMIZATION WITH PREDICTIONS: EXACT WORST CASES, QUERY HARDNESS, AND PER-RUN CERTIFICATES

Anonymous

ABSTRACT

1 INTRODUCTION

2 MODEL AND PRELIMINARIES

A ground set N with $|N| = n$, a monotone submodular $f : 2^N \to \mathbb{R}_{\geq 0}$ with $f(\emptyset) = 0$, and a cardinality budget K are fixed. The algorithm cannot evaluate f; it can only query a predictor $\tilde{f} : 2^N \to \mathbb{R}$ with $\tilde{f}(\emptyset) = 0$. Marginal gains are written $d_e(S) = f(S \cup \{e\}) - f(S)$ and $\tilde{d}_e(S) = \tilde{f}(S \cup \{e\}) - \tilde{f}(S)$. Throughout, $1 \leq K \leq n$; when $f(O^*) = 0$ every ratio statement is read as holding trivially, and $K \geq 1$ excludes empty runs.

Definition 1 (Prediction error). *For $\eta_u, \eta_o \geq 1$ the predictor $\tilde{f}$ has (single-element, multiplicative) error at most (η_u, η_o) if for every $S \subseteq N$ and $e \notin S$,*

$$d_e(S)/\eta_u \ \leq\ \tilde{d}_e(S) \ \leq\ \eta_o\, d_e(S).$$

In particular $d_e(S) = 0$ forces $\tilde{d}_e(S) = 0$, and all predicted gains are nonnegative. The scalar error is $\eta = \eta_u \eta_o$.

The multiplicative form of Definition 1 follows the error conventions of learning-augmented scheduling, where predicted weights or processing times are accurate up to multiplicative factors and guarantees degrade with that factor (Lattanzi et al., 2020; Azar et al., 2021; Zhao et al., 2022).

Only the product $\eta = \eta_u \eta_o$ matters for worst-case statements, in the following class sense.

Lemma 2 (Scaling). *Write $\mathcal{F}(\eta_u, \eta_o)$ for the set of predictors satisfying Definition 1 for a given monotone submodular f. For every $c \in [1/\eta_o,\ \eta_u]$ the map $\tilde{f} \mapsto c\tilde{f}$ is a bijection from $\mathcal{F}(\eta_u, \eta_o)$ onto $\mathcal{F}(\eta_u/c,\ c\,\eta_o)$, whose factors are again at least 1 and have the same product. The map preserves f and every comparison of predicted values, so each run of a comparison-based algorithm, predictive greedy in particular, and hence its worst-case ratio over the class, depends on (η_u, η_o) only through η.*

Approximation ratios are stated as $\alpha \in (0, 1]$ with $F^{\mathrm{ALG}} \geq \alpha\, F^{\mathrm{OPT}}$; larger is better.

Predictive greedy is the single-step greedy run on $\tilde{f}$: for $t = 0, \ldots, K-1$ it adds an element maximizing $\tilde{d}_e(S^t)$, with ties broken adversarially in all worst-case statements. The run always executes exactly K steps; a step whose maximum predicted gain is zero still selects, and the cases of Definition 3 cover it. A variant that stops early obtains none of the guarantees below: it is limited to the executed-steps product bound of Remark 25, and a two-element instance drives it to ratio $1/2$ against $L_2(1) = 3/4$. Its exact worst-case ratio at error level η is written $\rho_K(\eta)$. measures refine η:

Definition 3 (Selection error). *For an execution of predictive greedy with states $S^0, \ldots, S^{K-1}$ and picks $e_0, \ldots, e_{K-1}$, write $M_t = \max_{e \notin S^t} d_e(S^t)$ and $g_t = d_{e_t}(S^t)$ (monotonicity of f gives $g_t \geq 0$), and set*

$$a_t \ =\ \begin{cases} M_t/g_t, & g_t > 0, \\ 1, & M_t = g_t = 0, \\ \infty, & g_t = 0 < M_t. \end{cases} \qquad \eta^{\mathrm{sel}} \ =\ \max\{1, a_0, \ldots, a_{K-1}\},$$

the largest factor by which a step of the run fell short of the best true gain; a step whose chosen true gain vanishes while a better candidate exists makes the selection error infinite. Bounds evaluated at η^{sel} use the convention $L_K(\infty) = 0$. In the terminology of Goundan and Schulz, a run with finite η^{sel} is an execution of greedy with an α-approximate incremental oracle with $\alpha = \eta^{\mathrm{sel}}$; their condition is a per-step condition, which is what the case $a_t = \infty$ records (Goundan & Schulz, 2007).

The trajectory error restricts Definition 1 to the states of the run; it is used in a remark and in the appendix, and written η^{tr} when a symbol is needed. For every run of a predictor with finite error (η_u, η_o) the chain $\eta^{\mathrm{sel}} \leq \eta^{\mathrm{tr}} \leq \eta$ holds; in particular $a_t = \infty$ cannot occur then, because a vanishing chosen predicted gain forces every candidate's predicted, and then true, gain to vanish.

Table 2 in Appendix A collects the notation.

3 RELATED WORK

Approximate incremental oracles. The closest error model replaces the exact greedy step by an oracle returning an element whose true marginal gain is within a factor of the best available one, the α-approximate incremental oracle of Goundan & Schulz (2007), whose $\alpha \geq 1$ is exactly the selection error η^{sel} of Definition 3. Their Theorem 1 gives $1 - (1 - 1/(\alpha k))^k \geq 1 - e^{-1/\alpha}$ under such an oracle, which at $\alpha = 1$ is the classical cardinality-constrained bound of Nemhauser et al. (1978), whose asymptotic constant $1 - e^{-1}$ is unimprovable by polynomially many value queries (Nemhauser & Wolsey, 1978) (at finite K, partial enumeration trades a larger budget for a better constant); for the related set cover problem no polynomial-time algorithm approximates within $(1 - \varepsilon) \ln n$ unless $\mathrm{NP} \subseteq \mathrm{DTIME}(n^{O(\log \log n)})$ (Feige, 1998). We differ in the question, not the bound: Proposition 6 is their theorem restated in the model of Section 2 and attributed as such, while what is new is the exact worst-case ratio at every error level (Theorem 10), a per-K attaining family at error η^{sel} (Theorem 8), and an asymptotically matching bounded-query obstruction (Theorem 19).

Approximate submodularity. A second line weakens the objective instead of the access, measuring distance from submodularity by the submodularity ratio γ of Das & Kempe (2011, Definition 2.3). Greedy then still returns a $(1 - e^{-\gamma})$ fraction of the optimum (Das & Kempe, 2011, Theorem 3.2), and Elenberg et al. (2018) derive weak submodularity from restricted strong convexity, extending constant-factor greedy guarantees beyond submodular objectives. There the oracle is exact and the structure approximate, here the structure is assumed and the access is not; the third experiment family of Section 5 lies in the intersection, its true objective being measurably neither monotone nor submodular.

Noise and inexact oracles. A third line perturbs the value oracle itself, by random noise or by allowing the queried function to be only approximately submodular. Hassidim & Singer (2017) recover a ratio arbitrarily close to $1 - 1/e$ under independent noise once the budget is large enough, and state that adversarial noise admits no non-trivial guarantee; Horel & Singer (2016, Theorem 3) show that for ε-approximately submodular functions with ε above roughly $n^{-1/2}$ no algorithm with polynomially many queries beats $O(n^{-\delta})$; and Bhawalkar et al. (2025) treat a persistent noisy value oracle, where repeated queries to a set return one value, obtaining $1 - 1/e$ for monotone objectives under a matroid constraint, $1/e$ in the non-monotone matroid case and $1/2$ unconstrained. Our error is a deterministic multiplicative band on marginal gains, adversarial inside the band, so the answer is a finite ratio at every error level rather than either full recovery or vacuity, and the selection error we state it at is a property of the finished run.

Learning-augmented algorithms and learn-then-optimize. This framework asks for algorithms that improve with better predictions but do not degrade much when the predictions are poor (Purohit et al., 2018), the first property being consistency in its learning-augmented sense; Lykouris & Vassilvitskii (2021) realize it for caching with a competitive ratio that decreases as the oracle's error decreases and is capped at $O(\log k)$. Learning a submodular objective is itself hard, since PMAC-learning monotone submodular functions under arbitrary distributions forces approximation factor $\tilde{\Omega}(n^{1/3})$ (Balcan & Harvey, 2018), and against a background of negative results in which optimizing a learned surrogate lands arbitrarily far from the true optimum, Rosenfeld et al. (2018) characterize which surrogates can be optimized at all. Predicting other objects does help: a $1/2 - \varepsilon$ expected ratio for dynamic monotone submodular maximization under predicted insertion and deletion times, with amortized update time degrading in the prediction error η_{AB} (Agarwal & Balkanski, 2024),[1] and noisy predictions of the optimal solution bypassing classical hardness for problems such as max-cut (Cohen-Addad et al., 2024). What is predicted there is the solution, the schedule, or a function learned once and then optimized; here the prediction is the objective, queried adaptively during the run, and the guarantee is stated in an error measure the run reports.

Optimization from samples and query models. When only sampled sets and values are available, bounded curvature still permits a positive result (Balkanski et al., 2016), while in general no constant-factor approximation follows from polynomially many samples even for coverage functions learnable to arbitrary accuracy (Balkanski et al., 2017). We keep adaptive query access and move the difficulty into the accuracy of the answers, so Theorem 19 concerns what a bounded number of small queries extracts from an inaccurate predictor, not what a fixed sample reveals.

4 Theoretical results

4.1 An error bound is necessary

Proposition 4 (No bound, no guarantee)**.** *If no upper bound on η is assumed, then for every deterministic algorithm with arbitrary query access to $\tilde{f}$ and every $n \geq 2K$ there are pairs $(f, \tilde{f})$ on which the output T satisfies $f(T) \leq \frac{K}{n-K} f(O^*)$. Consequently no constant worst-case ratio is achievable without an error assumption.*

Proposition 4 says that some error bound is needed; the next statement says the bound has to be on marginal gains rather than on values. Following Hassidim and Singer, call $\tilde{f}$ value-accurate at level $\varepsilon \in (0, 1)$ if $(1-\varepsilon)f(S) \leq \tilde{f}(S) \leq (1+\varepsilon)f(S)$ for every $S \subseteq N$ (Hassidim & Singer, 2017).

Proposition 5 (Value accuracy is neither sufficient nor necessary)**.**

(i) *For every $\varepsilon \in (0, 1)$ there are a monotone submodular f and a predictor $\tilde{f}$, value-accurate at level ε, with $\tilde{d}_e(S) = 0$ at a pair where $d_e(S) > 0$; hence no (η_u, η_o) of Definition 1 is finite for $\tilde{f}$, and no bound of the form $L_K(\eta)$ follows from value accuracy alone.*

(ii) *For every $M > 0$ and every monotone submodular f not identically zero, the predictor $\tilde{f} = (1 + M)f$ fails value accuracy at every level $\varepsilon < M$, yet predictive greedy on $\tilde{f}$ picks at every state an element of maximum true gain, so $\eta^{\mathrm{sel}} = 1$ and the full guarantee of Proposition 6 applies to the run.*

(iii) *Conversely, every predictor with error at most (η_u, η_o) for a nonnegative f is value-accurate at level $\max\{1 - 1/\eta_u,\ \eta_o - 1\}$, provided that this value is below 1 (that is, $\eta_o < 2$), so that it lies in the domain of the definition above.*

4.2 The guarantee, stated for the selection error

Proposition 6 (Guarantee of predictive greedy)**.** *Let f be monotone submodular with $f(\emptyset) = 0$ and let $T = S^K$ be the output of a run of predictive greedy whose selection error is η^{sel} (Definition 3, with $L_K(\infty) = 0$). Then, with $L_K(x) = 1 - (1 - \frac{1}{xK})^K$,*

$$f(T) \;\geq\; L_K(\eta^{\mathrm{sel}})\, f(O^*) \;\geq\; \big(1 - e^{-1/\eta^{\mathrm{sel}}}\big) f(O^*).$$

If moreover the predictor has finite error (η_u, η_o), the same bound holds with η^{sel} replaced by η^{tr} or by η, and the three bounds are ordered $L_K(\eta^{\mathrm{sel}}) \geq L_K(\eta^{\mathrm{tr}}) \geq L_K(\eta)$.

Proposition 6 is restated in the prediction-error model of Section 2; it is essentially due to Goundan and Schulz (Goundan & Schulz, 2007, Theorem 1). A proof is included in Appendix B.3 for completeness.

[1] Their η_{AB} counts elements whose actual insertion or deletion time is far from the predicted one; it is unrelated to the multiplicative marginal-gain error η of Definition 1 and is renamed here to keep the two apart.

Remark 7. *η^{sel} is computable from the run when f becomes observable in hindsight, which makes Proposition 6 a certificate: the measured η^{sel} of a finished run certifies the printed bound for that run, and a run with a harmful zero step ($a_t = \infty$) certifies nothing, which the value $L_K(\infty) = 0$ makes explicit. The certificate reading requires the true objective to be monotone submodular; on objectives outside the model the same measurement is only a selection diagnostic. The experiments of Section 5 report which of their objectives support certificates and which do not.*

4.3 PER-K TIGHTNESS

Theorem 8 (Tightness for every K)**.** *For every $K \geq 2$ and $\hat{a} > 1$ there is an instance $(f, \tilde{f})$ on $2K$ elements with $\eta^{\mathrm{sel}} = \eta^{\mathrm{tr}} = \hat{a}$ on the adversarial-tie run of predictive greedy, whose output value is exactly $L_K(\hat{a})\, f(O^*)$. Hence Proposition 6 is an equality on this family for every K, not only in the limit.*

4.4 THE COHERENCE LEMMA

Lemma 9 (Coherence)**.** *Let f be monotone, $S \subseteq N$, $e, e' \notin S$, and suppose $\tilde{d}_e(S) \geq \tilde{d}_{e'}(S)$. Then*

$$\text{(i)}\ \ d_e(S \cup \{e'\}) \geq \tfrac{1}{\eta}\, d_{e'}(S \cup \{e\}), \qquad \text{(ii)}\ \ \big(1 - \tfrac{1}{\eta}\big)\, d_{e'}(S \cup \{e\}) \geq d_{e'}(S) - d_e(S).$$

Part (ii) has a sharp form that is the starting point of the proof of the next theorem. Let e be the element the predictor prefers at S and e' a competing candidate, and write $d = d_e(S)$ for the chosen true gain, $g = d_{e'}(S)$ for the competitor's current true gain and $h = d_{e'}(S \cup \{e\})$ for the same competitor's gain after the step; then (ii), combined with submodularity of f, reads

$$d - \frac{g}{\eta} \ \geq\ \Big(1 - \frac{1}{\eta}\Big)(g - h) \ \geq\ 0.$$

The left-hand side is the margin by which the chosen gain exceeds the smallest value the error band allows it, and the middle expression is the amount by which the step erodes the competitor's next gain, so the two kinds of damage cannot both be extreme at the same step. In particular, for $\eta > 1$ a step with $d = g/\eta$ forces $h = g$: a maximally wrong selection cannot also erode that candidate's future gain.

4.5 THE EXACT WORST CASE

For $K \geq 2$, $\eta \geq 1$, let $k_1 = (K-1)\eta + 1$, $q = (K-1)\eta/k_1$, and for $0 \leq j \leq K-1$

$$V_j(\eta) \ = \ 1 - q^j \Big(1 - \frac{K-j}{K\eta}\Big).$$

Theorem 10 (Exact worst-case ratio of predictive greedy)**.** *For every $K \geq 2$ and $\eta \geq 1$, under adversarial tie breaking,*

$$\rho_K(\eta) \ = \ \min_{0 \leq j \leq K-1} V_j(\eta),$$

and the minimum is attained by V_j on the segment $\eta \in [K-j, K-j+1]$ (with $V_0 = 1/\eta$ on $[K, \infty)$), so the breakpoints are the integers $2, \dots, K$. In particular $\rho_K(\eta) = 1/\eta$ exactly when $\eta \geq K$, and for $K \in \{2,3,4\}$ the closed forms are $\rho_2 = \min\{\frac{1}{\eta}, \frac{3}{2(\eta+1)}\}$, $\rho_3 = \frac{16\eta+3}{3(2\eta+1)^2}, \frac{7}{3(2\eta+1)}, \frac{1}{\eta}$ on $[1,2], [2,3], [3,\infty)$, and $\rho_4 = \frac{135\eta^2+36\eta+4}{4(3\eta+1)^3}, \frac{21\eta+2}{2(3\eta+1)^2}, \frac{13}{4(3\eta+1)}, \frac{1}{\eta}$ on $[1,2], [2,3], [3,4], [4,\infty)$.

Remark 11 (Ground-set size)**.** *Write $\rho_{n,K}(\eta)$ for the worst case over instances on a fixed ground set of size n. Restricting a worst-case instance to $T \cup O^*$ leaves the run legal and the ratio unchanged, and padding with elements of identically zero true and predicted gain preserves the band and the run, so $\rho_{n,K}(\eta) = \rho_{2K,K}(\eta)$ for every $n \geq 2K$; ρ_K denotes this common value throughout.*

How the two directions fit together. The starting point is the sharp form of Lemma 9: at a step where the run does not take a best candidate, the loss incurred now and the erosion of that candidate's later gain are linked, so an adversary cannot send both to their extremes at the same step. What the limit permits is a two-phase trajectory, a geometric phase in which the run buys gain q^t/k_1 while the marginals of the optimal elements decay by the factor q, followed by a frozen phase in which the chosen gain sits at the bottom of the band and those marginals stop decaying, the switch occurring after j steps. The lower bound charges that shape to every run at once: a fixed nonnegative combination of the four constraint families of Appendix B.6 equals $\sum_{t<K} d_t - V_j$ identically, so no run outputs less than V_j on the segment of j, and the explicit instances of the same appendix attain the value. Proposition 12 below reads that combination backwards: on the interior of a segment the equality case pins the two-phase trajectory exactly, so its shape is forced by the constraints rather than chosen by a construction. The integer breakpoints are where the forcing lapses, because the multipliers that carry it vanish at the segment endpoints (Remark 13).

For a run with picks $e_0, \ldots, e_{K-1}$ and states $S^0 = \emptyset, \ldots, S^K$ and for an enumeration $O^* = \{o_1, \ldots, o_K\}$, write $d_t = d_{e_t}(S^t)$, and let $g_{t,i}$ be $d_{o_i}(S^t)$ when $o_i \notin S^t$ and 0 otherwise. These are the variables of the reduced linear program of Appendix B.6.

Proposition 12 (Structure of worst-case runs)**.** *Let $K \geq 2$, normalize $f(O^*) = 1$, and let either $\eta \in (K-j, K-j+1)$ for some j with $1 \leq j \leq K-1$, or $j = 0$ and $\eta > K$. If an adversarial-tie run of predictive greedy attains $f(S^K) = \rho_K(\eta) = V_j(\eta)$, then the variables it induces are*

$$d_t = \begin{cases} q^t/k_1, & t < j, \\ q^j/(K\eta), & t \geq j, \end{cases} \qquad g_{t,i} = \frac{q^{\min(t,j)}}{K} \quad \text{for } 0 \leq t \leq K \text{ and every } i.$$

The conclusion constrains the chosen gains and the marginals of the optimal elements along the run. It does not determine the pair $(f, \tilde{f})$, and it does not constrain the marginals of the candidates the run does not select. Since every $g_{t,i}$ above is positive, while the convention of Appendix B.13 sets $g_{t,i} = 0$ once o_i has been selected, a run attaining the exact value in the stated range is disjoint from the fixed optimal set O^*.

Remark 13 (Breakpoints as an active-constraint switch)**.** *The integers at which the minimizing index changes are the error levels where the multipliers carrying Proposition 12 degenerate. In the certificate of Appendix B.6 the prediction multiplier $\lambda_P(j) = \big(\eta - (K-j)\big)/\big(K(\eta-1)\big)$ vanishes at the left endpoint of the segment of j and the coverage multiplier $\lambda_S(j) = (K-j+1-\eta)/k_1$ vanishes at its right endpoint, so at an integer η fewer constraints are forced and the argument of the proposition no longer applies. This is a switch of the active constraint set, and the uniqueness does not extend across it: at $K=3$ and $\eta = 2$ the two chosen-gain sequences $(1/5,\ 2/15,\ 2/15)$ and $(1/5,\ 4/25,\ 8/75)$, coming from the $j=1$ and the $j=2$ instances of Appendix B.6, both sum to $7/15 = \rho_3(2)$.*

Remark 14 (Ties, and what near-equality forces)**.** *In the range of Proposition 12, attaining the exact value forces a predicted-gain tie with an optimal element at every step. The two constraints of the reduced program that carry the error band, the coherence constraint $\mathrm{cons}(t,i)$ and the greedy-choice constraint $\mathrm{pred}(t,i)$ of Appendix B.13, have slacks that decompose into three nonnegative terms each (Appendix B.13 and Appendix B.7), whose middle term in both cases is $\big(\tilde{d}_{e_t}(S^t) - \tilde{d}_{o_i}(S^t)\big)/\eta_o$; the proposition sets both slacks to zero, the first for $t \leq K-2$ and the second at $t = K-1$, so the predicted gain of the selected element equals that of every optimal element at every step. This is why adversarial tie breaking is a hypothesis of Theorem 10 and not a convenience. The same certificate also controls near-equality: a run whose output is $V_j(\eta)+\varepsilon$ has $0 \leq \mathrm{slack}_r \leq \varepsilon/\lambda_r$ for every constraint r whose multiplier λ_r is positive, since the weighted sum of the slacks equals ε. That is a bound checkable one run at a time and not a uniform stability statement: turning it into a distance bound on the whole trajectory would have to accumulate the errors of the recursion above and to handle the multipliers degenerating as η approaches a breakpoint or as K grows.*

Remark 15. *L_K and ρ_K differ by $O(1/K)$ for fixed η, and the previously used explicit family attains only $U_K(\eta) = 1 - (1 - \frac{1}{\eta(K-1)+1})^K$, which is strictly above V_{K-1} for every $\eta > 1$. The predictor $\tilde{f}$ of the attaining instances is monotone but not submodular. Requiring the surrogate to be submodular changes the worst case: at $\eta = 1$ the band forces $\tilde{f} = f$ and the two models coincide, while at the verified points with $1 < \eta < K-1$ the worst case strictly improves; at $K=3$, $\eta = 1.5$ the value rises from $9/16$ to $19/33$, and at $K=4$, $\eta = 2$ from $22/49$ to $23/50$. The family above becomes inadmissible in that model, so the upper-bound argument through U_K no longer applies there; at the verified points U_K itself still lies above the new worst case. A general characterization is left open.*

4.6 THE $1/\eta$ CEILING

Theorem 16 (Ceiling at $1/\eta$). *For every deterministic algorithm with arbitrary query access to $\tilde{f}$, every $\eta_u, \eta_o \geq 1$, and every $n \geq 2K$, there is a pair $(f, \tilde{f})$ with error exactly (η_u, η_o) on which the output T satisfies $f(T) \leq f(O^*)/\eta$. For every randomized algorithm there is a fixed pair, again with error exactly (η_u, η_o), on which the expectation over the algorithm's randomness satisfies $\mathbb{E}[f(T)] \leq \big((1 - \frac{K}{n})\frac{1}{\eta} + \frac{K}{n}\big) f(O^*)$. Conversely, exhaustive search over all K-subsets of predicted values achieves $f(\hat{S}) \geq f(O^*)/\eta$ on every instance.*

The randomized expression is an upper bound on the randomized worst case; it is not claimed to be the exact randomized value at finite n. For $n < 2K$ the same symmetric construction gives the sharper deterministic ceiling $K/\big((2K-n)+(n-K)\eta\big)$, which interpolates between $1/\eta$ at $n = 2K$ and 1 at $n = K$ and is attained by exhaustive search on the LP-verified grid $K \leq 4$, $n \leq 7$ (general attainment is a conjecture).

Remark 17 (Exhaustive search against greedy, by worst-case guarantee). *For $1 \leq \eta < K$ the worst-case guarantee $1/\eta$ of exhaustive search over predicted values is strictly better than the exact worst case of predictive greedy: $\rho_K(\eta) \leq V_1(\eta) < V_0(\eta) = 1/\eta$, with $V_0 - V_1 = (K-\eta)/(K\eta k_1)$. At $K = 3$ the pairs (greedy, exhaustive) are $(19/27,\ 1)$ at $\eta = 1$, $(9/16,\ 2/3)$ at $\eta = 3/2$, $(7/15,\ 1/2)$ at $\eta = 2$, and both equal $1/3$ at $\eta = 3$. This compares worst-case guarantees and does not say that the exhaustive output has larger true value on every instance; the exhaustive algorithm also evaluates $\binom{n}{K}$ predicted sets. For $\eta \geq K$, $\rho_K(\eta) = 1/\eta$ (Theorem 10), so there predictive greedy itself attains the unbounded-query deterministic optimum.*

4.7 ASYMPTOTICS

Corollary 18 (Limit in K). *For fixed $\eta \geq 1$, both $L_K(\eta)$ and $\rho_K(\eta)$ converge to $1 - e^{-1/\eta}$ as $K \to \infty$; L_K is monotone in K, and ρ_K is non-increasing in K, equal to $1/\eta$ for $K \leq \lfloor \eta \rfloor$ and strictly decreasing from $K \geq \lfloor \eta \rfloor$ on, so the limit is approached from above.*

Quantitatively, for fixed $\eta \geq 1$, $\rho_K(\eta) = 1 - e^{-1/\eta} + c(\eta)/K + O(1/K^2)$ with $c(\eta) = e^{-1/\eta}(2\eta - 1)/(2\eta^2)$; the same $1/K$ coefficient is shared by U_K, which is why $U_K - \rho_K = O(1/K^2)$, while L_K has $1/K$ coefficient $e^{-1/\eta}/(2\eta^2)$.

4.8 HARDNESS FOR BOUNDED-QUERY ALGORITHMS

The ceiling of Theorem 16 places no restriction on the query budget. The final result bounds instead what a budget of small queries extracts at a prescribed error level. Fix an integer τ with $1 \leq \tau < K$ and a design parameter $\theta \geq 1$, and put

$$a_\theta = 1 - \frac{1}{\theta K}, \qquad A_\theta = a_\theta^\tau \, \frac{K}{K-\tau}, \qquad B_\theta = a_\theta^{1-\tau},$$

$$\delta(\theta) = A_\theta B_\theta - 1 = \frac{\tau - 1/\theta}{K-\tau}, \qquad \Psi(\theta) = \theta\big(1 + \delta(\theta)\big) = \frac{\theta K - 1}{K - \tau}.$$

The pair built from θ in Appendix B.10 has smallest admissible error factors $\sqrt{\theta}A_\theta$ and $\sqrt{\theta}B_\theta$, whose product is $\Psi(\theta)$, and Ψ inverts in closed form: the design parameter that realizes a prescribed scalar error η is $\bar{\theta} = \Psi^{-1}(\eta) = \big(\eta(K-\tau)+1\big)/K$, which is at least 1 exactly when $\eta \geq (K-1)/(K-\tau)$. The quantity $\Phi(\theta) = \theta \max\{A_\theta, B_\theta\}^2$ and its inverse $\hat{\eta} = \Phi^{-1}(\eta)$ calibrate the same family under the extra requirement $\eta_u = \eta_o$. Since $A_\theta B_\theta \leq \max\{A_\theta, B_\theta\}^2$, with equality only at $A_\theta = B_\theta$, the value $\Phi(\theta)$ is an admissible error bound for the pair but is in general strictly above the pair's own error, so the calibration through $\bar{\theta}$ is the one used below.

Theorem 19 (Bounded-query hardness, deterministic). *Let $c \geq 0$ be real and put $\tau = \lceil c \rceil + 1$, which equals $c + 1$ at integer c. Let the integers $K > \tau$ and $n \geq 4K^{c+2}$ and the real $\eta > 1$ with $\eta \geq \frac{K-1}{K-\tau}$ be fixed, so that $\bar{\theta} \geq 1$. For every deterministic algorithm making at most n^c queries to $\tilde{f}$, each on a set of size at most K, there is an instance $(f, \tilde{f})$ with monotone submodular f, whose smallest admissible error factors in Definition 1 have product exactly η, on which the output T (of* *size at most K) satisfies*

$$\frac{f(T)}{f(O^*)} \;\le\; H_{K,\tau}(\eta) \;:=\; 1-\Big(1-\frac{1}{\eta(K-\tau)+1}\Big)^{K} \;=\; L_K(\bar{\theta}).$$

Any prescribed split $\eta_u\eta_o = \eta$ of that error is realized by the rescaling of Appendix B.10. For randomized algorithms the same bound holds in expectation up to an additive

$$\varepsilon_n = \frac{K}{n} + \frac{K^{2\tau+2}}{(\tau+1)!\, n^{\tau+1-c}},$$

which for integer c reads $\frac{K}{n} + \frac{K^{2c+4}}{(c+2)!\, n^2}$. As $K\to\infty$ with τ and θ fixed, $K\delta(\theta)\to\tau-1/\theta$; with c and η fixed, $H_{K,\tau}(\eta)\to 1-e^{-1/\eta}$.

Three scope notes. The additive term of the randomized bound does not vanish on its own: at the smallest permitted $n = 4K^{c+2}$ and integer c its second summand equals $1/(16\,(c+2)!)$, a constant, so a limit statement for randomized algorithms selects $n(K)$ with $\varepsilon_n\to 0$, for example $n \ge 4K^{c+3}$. The cap on query size is a condition of the query model: it neither follows from nor implies polynomial running time, and computation between queries is unrestricted. At finite parameters $H_{K,\tau}$ can exceed the ceiling $1/\eta$ of Theorem 16 (at $\eta = 2$, $c = 2$, $K = 8$ it lies above $1/2$), so the two upper bounds combine as $\min\{H_{K,\tau}(\eta),\, 1/\eta\}$; the content of Theorem 19 is the asymptotic match with L_K, not a pointwise improvement.

Remark 20. *At $\tau = 1$, which under $\tau = \lceil c\rceil + 1$ means $c = 0$, the hardness constant is the value of the explicit family of Section 4.3: $H_{K,1}(\eta) = U_K(\eta)$. For every $K\ge 2$ and $\eta > 1$ the four curves are then ordered*

$$L_K(\eta) \;\le\; \rho_K(\eta) \;\le\; U_K(\eta) \;=\; H_{K,1}(\eta),$$

that is, the guarantee of Proposition 6, the exact worst case of predictive greedy (Theorem 10), the value of the explicit family (Remark 15), and the ceiling that Theorem 19 places on algorithms allowed a single query of size at most K. The second inequality is strict for $\eta > 1$.

Remark 21. *Combining Proposition 6 with Theorem 19 pins the bounded-query optimum at error level η to $1-e^{-1/\eta}$ asymptotically in K, for query classes that contain predictive greedy. The direct implementation of predictive greedy evaluates $\tilde{f}$ on at most $Kn - K(K-1)/2$ sets, one cached value per state and one query per remaining candidate, rather than on K sets per step, so $nK \le n^c$ is a sufficient condition; integer $c \ge 2$ with $K \le n$ is one regime in which it holds. The qualifier cannot be dropped. At $c = 0$ a deterministic algorithm allowed a single query of size at most K can be forced to output a set of value zero. Its query Q and, after the answer, its output T are fixed sets; for $n \ge 2K+2$, place two further elements outside $Q\cup T$, give them true modular weights 1 and 1 and predicted weights 1 and 2, and give every remaining element weight zero in both. The predictor then answers 0 on Q, which is the answer the run already received, so the run is unchanged, while the scalar error is exactly 2 and $f(T) = 0$. At error level 2 the worst-case ratio of that class is therefore 0 and not $1-e^{-1/2}$. This does not collide with Theorem 19, which caps the ratio from above and does not assert that the cap is attained. At finite K the gap between $L_K(\eta)$ and $H_{K,\tau}(\eta)$ is of order $(c+1)/K$; the form $O(c/K)$ would report that gap as zero at $c = 0$, where the same example shows it is not.*

Remark 21 measures budgets on the scale n^c, which is coarser than the budget predictive greedy actually spends. Re-running the counting argument of Theorem 19 at that exact budget gives a ceiling for the whole budget class, with the $\tau = 1$ constant of Remark 20 rather than the weaker constant that the scale $n^c \ge nK$ would force.

Corollary 22 (Optimality within the greedy query budget)**.** *Let $K\ge 2$, $\eta > 1$ and $n \ge 4K^5$, and let $\mathcal{A}_{\mathrm{lin}}$ be the class of deterministic algorithms that make at most nK queries to $\tilde{f}$, each on a set of size at most K, and output a set of size at most K; predictive greedy, at $Kn - K(K-1)/2$ queries, belongs to $\mathcal{A}_{\mathrm{lin}}$. For every $A \in \mathcal{A}_{\mathrm{lin}}$ there is an instance $(f, \tilde{f})$ with monotone submodular f, whose smallest admissible error factors in Definition 1 have product exactly η, on which the output T satisfies*

$$\frac{f(T)}{f(O^*)} \;\le\; U_K(\eta) \;=\; H_{K,1}(\eta) \;=\; 1-\Big(1-\frac{1}{\eta(K-1)+1}\Big)^{K},$$

so that, together with Theorem 16, the worst-case ratio of A at error level η is at most $\min\{U_K(\eta),\, 1/\eta\}$. For every randomized algorithm with the same budget there is again an instance with error exactly η on which the expectation over the algorithm's randomness satisfies $\mathbb{E}\big[f(T)/f(O^*)\big] \le U_K(\eta) + \varepsilon_n$, where $\varepsilon_n = K^2/n + K^5/(2n)$.

Remark 23. *Every run of predictive greedy on an instance with error at most η achieves at least $\rho_K(\eta)$ (Theorem 10), and Corollary 22 caps every member of $\mathcal{A}_{\rm lin}$ at $\min\{U_K(\eta), 1/\eta\}$, so the best worst-case ratio achievable within the greedy query budget lies in the interval $[\rho_K(\eta), \min\{U_K(\eta), 1/\eta\}]$. For $\eta \ge K$ the interval collapses: $\rho_K(\eta) = 1/\eta$ by Theorem 10 and $1/\eta \le U_K(\eta)$ by the chain of Remark 20, so predictive greedy attains the ceiling of its own budget class exactly. For $1 < \eta < K$, at fixed η as $K \to \infty$ the width of the interval is*

$$U_K(\eta) - \rho_K(\eta) \;=\; \frac{c'(\eta)}{K^2} + O\Big(\frac{1}{K^3}\Big), \qquad c'(\eta) \;=\; e^{-1/\eta}\, \frac{\lfloor \eta \rfloor \big(2\eta - \lfloor \eta \rfloor - 1\big)}{2\eta^2} \;>\; 0,$$

so any improvement over predictive greedy within $\mathcal{A}_{\rm lin}$ is confined to a band of width $O(1/K^2)$; whether any of it can be realized at finite K is open.

5 Experiments

The experiments ask what Proposition 6 certifies on surrogates that were never built to satisfy Definition 1, and how far that certificate sits below what predictive greedy delivers. The certificate is read at the measured stepwise selection error $\eta^{\rm sel}$ of the finished run (Remark 7); it applies only where the true objective satisfies the model, which the second family does and the first and third measurably do not, so their selection numbers are reported as diagnostics rather than certificates. Three surrogate families are ordered by their distance from the model of Section 2: surrogates *learned* from data, surrogates that evaluate the same objective on a *partially observed* input, and *heuristic* surrogates for a true objective that itself lies outside the model. The instances constructed for Theorems 10 and 8 are run through the same pipeline as a control.

Setup. In every family f is the quantity of interest, $\tilde f$ is what the algorithm may query, and nothing synthetic is inserted between them. *(E1) Feature selection:* $f(S)$ is the held-out accuracy of a decision tree trained on the features S, and $\tilde f(S)$ the mean 5-fold cross-validation accuracy of the same estimator on the training split alone, so the held-out split is structurally out of the predictor's reach; 4 datasets (the largest with 25,375 rows), 30 splits each, $K = 1, \ldots, 7$. *(E2) Influence maximization:* f is one-hop coverage in a network and $\tilde f$ the same coverage on an observed graph keeping each edge independently with probability $p \in \{0.3, 0.5, 0.8\}$ (Kempe et al., 2003); 4 networks (largest 50,515 nodes, 819,306 edges), 20 observed graphs per network and p, giving 240 trajectories of length 30 (Rozemberczki et al., 2019; Leskovec et al., 2007; Yin et al., 2017). *(E3) Extractive summarization:* f is the ROUGE-1 F score against a reference summary and $\tilde f$ one of 3 standard submodular sentence-selection objectives that never see the reference (Lin & Bilmes, 2011); 299 articles in 3 categories of the BBC news corpus (Greene & Cunningham, 2006), $K = 3, \ldots, 7$. Ratios are $f(S^{\tilde f})/f(S^f)$, whose denominator is the value of greedy on the true objective, a feasible solution and hence a lower estimate of the optimum, so the printed ratio is an upper estimate of $f(S^{\tilde f})/f(O^*)$ (Table 1, note (i)); on the one dataset small enough to enumerate the optimum, that proxy inflates a ratio by 1.8 %. Greedy uses cached lazy evaluation; accelerated variants carry guarantees of their own (Mirzasoleiman et al., 2015) but are not used, because measuring $\eta^{\rm sel}$ needs the exact per-step maximizer of the true gain. Appendix C gives the protocol, the baseline panel and the enumeration.

Learned surrogates. Held-out accuracy is not a monotone submodular function of the feature set: 711 of the 720 budget prefixes observe a negative true candidate gain at a visited state, 208 select one, and 173 contain a harmful zero step, so for this family every selection number below is a diagnostic, not a certificate. At $K = 7$, over 120 runs, the median ratio is 0.971 (interquartile range 0.943 to 0.998) at median finite-step diagnostic 2.0, where the reference value $L_7(\eta^{\rm sel})$ is 0.405; on the largest dataset alone the ratio is 0.999 at diagnostic 1.55 (reference 0.492). Which measure a bound is read at decides whether it says anything: the trajectory error of the same runs has median at least 19.4 (the reported figure discards candidate pairs whose gains fall below one quantization

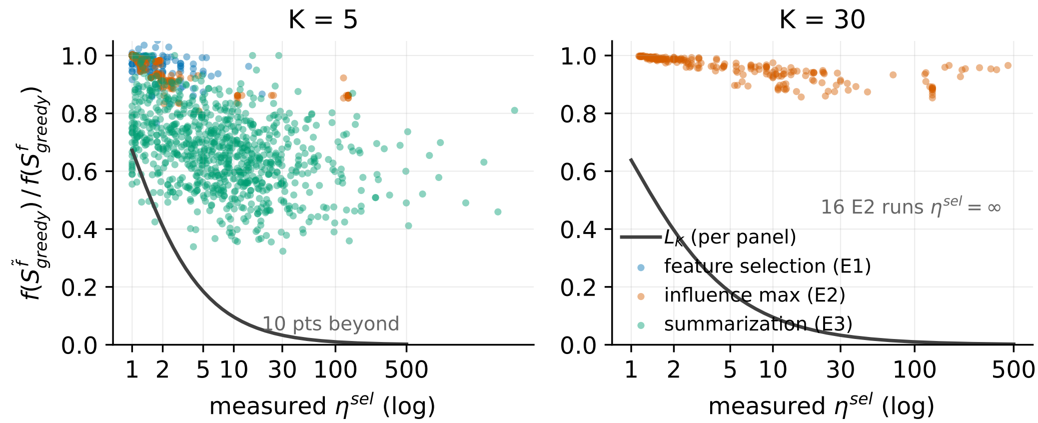

Figure 1: Measured ratio against the measured selection diagnostic η^{sel} (log scale) at $K = 5$ and $K = 30$, with the curve $L_K(\eta^{\text{sel}})$ of Proposition 6. The curve is a certificate only for runs whose true objective satisfies the model (influence maximization); for the other two families it is a reference value at their finite-step diagnostic. The 16 influence-maximization runs whose stepwise η^{sel} is infinite, so that their certified bound vanishes, are stated in the panel and not drawn. Ratio denominators are greedy on the true objective, a lower estimate of the optimum, so the plotted ratios are upper estimates.

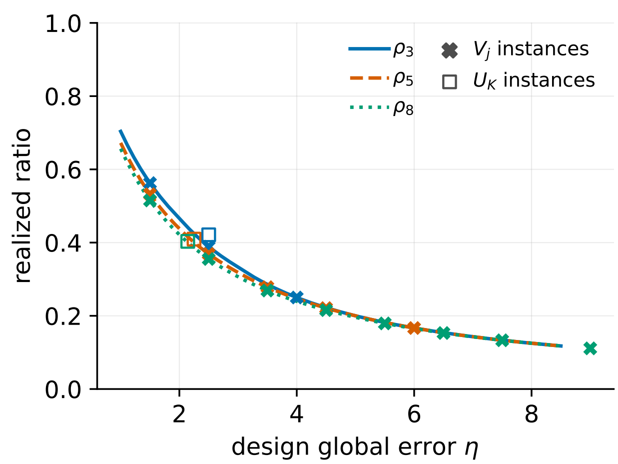

Figure 2: The exact worst case $\rho_K(\eta)$ of Theorem 10 on the global-error axis, with the 19 constructed instances executed by the experiment pipeline at their design errors ($K \in \{3, 5, 8\}$). Realized ratios match the theory to 1.1×10^{-16}. The global η of the real-task runs is not identified by the data, which is why they appear only in Figure 1.

unit of the accuracy, so it is a lower estimate; Appendix C), and at that value the same formula gives only 0.050.

Partially observed surrogates. Here f and $\tilde{f}$ are one coverage function of two nested edge sets, both monotone submodular, so Proposition 6 reads as a certificate. At $K = 30$ the median ratio is 0.963 (interquartile range 0.936 to 0.989) at median stepwise selection error 4.5, certifying $L_{30}(\eta^{\text{sel}}) = 0.199$ for the median run. Nested coverage guarantees that predicted and true VALUES never disagree in sign, but it does not align the zero sets of the marginal gains: 16 of the 240 trajectories (6.7 %) contain a step whose chosen true gain is zero while a positive candidate exists, their stepwise η^{sel} is infinite, and they certify nothing (Table 1, note (iii); 225 budget prefixes are affected in total).

Table 1: Predictive greedy across three surrogate families. Medians over all runs at the stated K (IQR in brackets). L_K is the certified lower bound $L_K(\eta^{\text{sel}})$ of Proposition 6 evaluated at the median measured η^{sel}; $d_t \leq 0$ % is the share of trajectory steps whose chosen true gain is non-positive.

Task (K)	Surrogate $\tilde{f}$	ratio	IQR	η^{sel}	L_K	sign-viol. %	$d_t \leq 0$ %
Feature sel. (7)	5-fold CV acc.	0.971	[0.943, 0.998]	2.0	0.405	22.7	21.4
Influence max. (30)	observed graph	0.963	[0.936, 0.989]	4.3	0.207	–	0.0
Summarization (5)	cover./div./FL	0.670	[0.576, 0.757]	7.2	0.132	10.0	20.0
Worst-case V_j (5)	constructed	0.278[a]	exact	3.5	ρ_K	0	0.0

[a] representative point $j = 2$, $\eta = 3.5$; every constructed instance realizes its theoretical value to 10^{-10} (results/E4_worst_case.csv).

Note (i): the ratio's denominator is greedy-on-f, a feasible solution and hence a lower estimate of OPT, so every ratio in this table is an upper estimate of $f(S^{\tilde{f}})/f(\mathrm{OPT})$.

Note (ii): the η^{sel} column is the finite-step ratio; a step with $d_t \leq 0$ and a positive candidate makes the stepwise η^{sel} infinite (certificate 0), and the last column reports how often such steps occur (column `frac_steps_nonpos` of the row CSVs); for the out-of-model families the column is a diagnostic, not a certificate.

Note (iii): for influence maximization f and $\tilde{f}$ are coverage functions on nested edge sets, so strictly opposite-sign $(d, \tilde{d})$ pairs cannot occur and the entry is "–"; zero-positive mismatches can occur and do (the harmful zero steps counted in the text).

Observation probability against error. This family lets the amount of error be dialled. Lowering p from 0.8 to 0.3 raises the median stepwise η^{sel} monotonically on 3 of the 4 networks, by factors from 3.2 to 88 between the endpoints; on the remaining network one intermediate setting has an infinite median (1 network-p cell, where half the runs contain a harmful zero step), which is the strongest form the degradation takes. The realized ratio degrades far more slowly over the same range, monotonically on all 4 networks, from no worse than 0.988 down to between 0.882 and 0.963. The certified bound moves fastest of the three: $L_{30}(\eta^{\text{sel}})$ falls from the range 0.361 to 0.573 down to 0.008 to 0.230, by up to a factor 64 on one network, and vanishes outright on the infinite-median cell. Missing observations therefore cost much less realized value than they cost certificate.

Heuristic surrogates outside the model. At $K = 5$, over 879 runs, the median ratio falls to 0.670 (interquartile range 0.576 to 0.757) at median finite-step diagnostic 7.2, where the reference value $L_5(\eta^{\text{sel}})$ is 0.132; the best single surrogate, the coverage objective, reaches 0.712. Across the three families the median selection error grows and the median ratio falls together, which is the spectrum Table 1 summarizes.

An out-of-model finding. The true objective of the third family does not satisfy the hypotheses of Proposition 6, and we measure by how much rather than leaving it implicit. On 70,560 triples $(A,\ A \cup \{b\},\ e)$ drawn from the articles with $|A| \leq 3$, the ROUGE-1 F score violates submodularity on 2.14 % and monotonicity on 7.12 % of them, the largest submodularity violation being 0.020 in units of the score, while the 3 surrogates record 0 violations of either property on the same triples; a further 20.0 % of trajectory steps have a nonpositive chosen true gain, which under Definition 3 would already send the stepwise η^{sel} to infinity, so the printed diagnostic is the finite-step ratio (Table 1, note (ii)). For this family the L_K column is therefore a reference value computed by the same formula and not a certificate, and recovering one would need a guarantee stated for approximately submodular objectives (Section 3).

The worst case, executed by the same code. 19 constructed instances were run through the pipeline that produced every other point: 16 instances of the V_j family of Theorem 10, one per segment at $K \in \{3, 5, 8\}$, and 3 of the per-K family of Theorem 8. Realized ratios match the theoretical values to 1.1×10^{-16}, and on the V_j instances the measured selection error matches the design error to 5.3×10^{-15}. The worst case is thus not an asymptotic figure of speech: it occurs point by point in the same implementation of greedy that produced the scatter of Figure 1, and Figure 2 shows the constructed instances on the axis where their design error lives. Three auxiliary figures, the distribution of η^{sel} by budget, the observation probability sweep, and the true against predicted gains with an error band drawn in, are deferred to Appendix C.

6 Conclusion

AI use statement

In this work, we used generative AI tools as research assistants: to write and run the LP, symbolic, and exhaustive verification scripts that back the status annotations of the theorems, to implement and execute the experiment pipelines, to draft theorem statements and proof text, and to organize data and references. We have reviewed all AI-assisted work: the authors re-ran the verification scripts and checked the proofs line by line before submission, and statements whose proofs have not been so checked are explicitly marked in the source with their verification status. No claim in this paper rests solely on an unverified AI-generated argument. We take responsibility for the final content of this work, including text, claims, and artifacts produced with the aid of generative AI.

Reproducibility statement

All numerical claims are backed by one-command scripts in the anonymous repository accompanying this submission (link withheld in this draft; an anonymized link is to be added at submission time): the factor-revealing and reduced LPs, the symbolic verifications (sympy, exact arithmetic), the worst-case instance realizations, and the three experiment pipelines with fixed seed lists. The appendix states, for each theorem, which parts are verified by which script. Data handling is documented in the repository inventory: the airline table and the BBC corpus ship with the repository; the reference summaries missing for two BBC categories are backfilled from a public CSV of the same dataset, with a token-level cross-validation of the backfill on the category present locally; the social graphs are public SNAP datasets used as replacements for graphs no longer available, as documented in the inventory. Experiment tables state that ratio denominators use the true-value greedy as an OPT proxy, and every reported lower bound states the error measure it is evaluated at.

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A Notation

B Proofs

Three conventions are in force throughout this appendix. First, f is monotone submodular with $f(\emptyset) = 0$ and $\tilde{f}(\emptyset) = 0$, and submodularity is used in its diminishing-returns form: $d_e(T) \le d_e(S)$ for $S \subseteq T$ and $e \notin T$. Second, an optimal set is written O^*, $|O^*| \le K$, and instances are normalized so that $f(O^*) = 1$ whenever a ratio is computed. Third, the tie-breaking rule of a greedy run is part of the instance: "adversarial ties" means that when several elements maximize the predicted gain, the one named by the construction is taken.

Two of the constructions below are counting functions, so the following observation is recorded once and used twice.

Lemma 24 (counting functions)**.** *Let $N = N_1 \cup \cdots \cup N_r$ be a partition and let $h(S) = H(|S \cap N_1|, \ldots, |S \cap N_r|)$ for a function H on the box $\prod_i \{0, 1, \ldots, |N_i|\}$. Write $\Delta_i H(v) = H(v + u_i) - H(v)$ for the forward difference in coordinate i (u_i the ith unit vector), defined when $v_i < |N_i|$. If $\Delta_i H \ge 0$ for every i and every v, and $\Delta_\ell \Delta_i H \le 0$ for every pair (i, ℓ), including $i = \ell$, and every v at which both differences are defined, then h is monotone and submodular.*

Argument. Let $v(S)$ denote the count vector of S. For $e \in N_i$ with $e \notin S$ one has $d_e(S) = \Delta_i H(v(S))$, which is nonnegative by hypothesis, so h is monotone. For $S \subseteq T$ and $e \in N_i \setminus T$, the vectors satisfy $v(S) \le v(T)$ coordinatewise and $v(T)_i < |N_i|$. Walk from $v(S)$ to $v(T)$ by single unit steps, raising coordinate i last: at each step the quantity $\Delta_i H$ changes by one second

Table 2: Notation used throughout the paper.

N, n	ground set and its size
K	cardinality budget
f	monotone submodular objective, $f(\emptyset) = 0$; not queryable
$\tilde{f}$	predictor of f; the only queryable object
$d_e(S)$, $\tilde{d}_e(S)$	true and predicted marginal gain of e at S
η_u, η_o	under- and over-prediction factors ($d/\eta_u \le \tilde{d} \le \eta_o d$ on single-element gains)
$\eta = \eta_u \eta_o$	scalar (global) prediction error
η^{sel}	selection error of a run (Definition 3); main text uses this
η^{tr}	trajectory error (error bound restricted to the run's states); one remark only
O^*, F^{OPT}	an optimal K-set and its value
$\alpha \in (0, 1]$	approximation ratio convention $F^{\text{ALG}} \ge \alpha F^{\text{OPT}}$
$\rho_K(\eta)$	exact worst-case ratio of predictive greedy, adversarial ties
$L_K(x) = 1 - (1 - \frac{1}{xK})^K$	guarantee curve (Proposition 6)
$k_1 = (K-1)\eta + 1$, $q = (K-1)\eta/k_1$	segment parameters of the exact worst case
$V_j(\eta) = 1 - q^j(1 - \frac{K-j}{K\eta})$	segment values; $\rho_K = \min_j V_j$ (Theorem 10)
$U_K(\eta) = 1 - (1 - \frac{1}{\eta(K-1)+1})^K$	value of the per-K tightness family at global error η
a_θ, $\delta(\theta)$, Ψ, $\bar{\theta}$	hardness-family parameters (Section 4.8); Φ, $\hat{\eta}$ are the superseded symmetric calib
$\tau = \lceil c \rceil + 1$, $Q = n^c$	balancedness margin and query budget of the hardness theorem ($\tau = c + 1$ at integ
ratio (experiments)	$f(S^{\tilde{f}}_{\text{greedy}})/f(S^{f}_{\text{greedy}})$; the denominator is an OPT proxy

difference $\Delta_\ell \Delta_i H \le 0$, and every intermediate vector has ith coordinate at most $v(T)_i < |N_i|$, so $\Delta_i H$ is defined all along. Hence $d_e(T) = \Delta_i H(v(T)) \le \Delta_i H(v(S)) = d_e(S)$, which is the diminishing-returns form of submodularity.

B.1 Necessity of an error bound (Proposition 4)

Fix $K \ge 1$ and $n \ge 2K$, and let $\mathcal{A}$ be an algorithm with arbitrary query access to $\tilde{f}$. Put

$$\gamma = \frac{K^2}{n(n-K)} \in (0, 1], \qquad \tilde{f}(S) = |S|,$$

where $\gamma \le 1$ because $n \ge 2K$ gives $n(n-K) \ge 2K \cdot K > K^2$. For a K-set $O \subseteq N$ define

$$f_O(S) = |S \cap O| + \gamma\,|S \setminus O|.$$

The pair is admissible and has finite error. f_O is modular with coefficients in $\{1, \gamma\} \subseteq (0, 1]$, so it is monotone and submodular with $f_O(\emptyset) = 0$, and $\tilde{f}(\emptyset) = 0$. Every single-element gain is $\tilde{d}_e(S) = 1$ and $d_e(S) \in \{1, \gamma\}$, so $\eta_u = \max_{S,e} d/\tilde{d} = 1$ and $\eta_o = \max_{S,e} \tilde{d}/d = 1/\gamma$: the error is exactly $\eta = 1/\gamma = n(n-K)/K^2$, which is finite for every n and grows without bound as $n \to \infty$. Also $\max_{|S| \le K} f_O(S) = f_O(O) = K$, because $\gamma \le 1$ makes the K largest modular coefficients exactly those of O.

Deterministic algorithms. $\tilde{f}$ does not depend on O, so the whole query transcript of $\mathcal{A}$, and therefore its output T with $|T| \le K$, are the same for every O. Since $n \ge 2K$ and $|T| \le K$, a K-set $O \subseteq N \setminus T$ exists. For that O, $T \cap O = \emptyset$ gives

$$\frac{f_O(T)}{f_O(O^*)} \;=\; \frac{\gamma|T|}{K} \;\le\; \gamma \;=\; \frac{K^2}{n(n-K)} \;\le\; \frac{K}{n-K}.$$

Randomized algorithms. Let O be a uniformly random K-subset, independent of the random string of $\mathcal{A}$. The transcript still does not depend on O, so the output T is independent of O, and the hypergeometric mean gives $\mathbb{E}\big[|T \cap O|\big] = \mathbb{E}|T| \cdot K/n \le K^2/n$. Bounding $f_O(T) \le |T \cap O| + \gamma K$ and dividing by $f_O(O^*) = K$,

$$\mathbb{E}\Big[\frac{f_O(T)}{f_O(O^*)}\Big] \;\le\; \frac{K}{n} + \gamma \;=\; \frac{K}{n} + \frac{K^2}{n(n-K)} \;=\; \frac{K(n-K) + K^2}{n(n-K)} \;=\; \frac{K}{n-K},$$

and $\min_O \leq \mathbb{E}_O$ produces a single O attaining the bound in expectation over the random string. Letting $n \to \infty$ with K fixed drives the right side to 0, so no constant worst-case ratio survives when η is unconstrained.

No script checks the computations of this proof.

B.2 VALUE ACCURACY VERSUS GAIN ACCURACY (PROPOSITION 5)

Part (i). Fix $\varepsilon \in (0, 1)$ and take $N = \{a, b\}$ with

$$f(\emptyset) = 0, \quad f(\{a\}) = 1, \quad f(\{b\}) = \tfrac{2\varepsilon}{1-\varepsilon}, \quad f(\{a, b\}) = \tfrac{1+\varepsilon}{1-\varepsilon},$$

$$\tilde{f}(\emptyset) = 0, \quad \tilde{f}(\{a\}) = 1 + \varepsilon, \quad \tilde{f}(\{b\}) = \tfrac{2\varepsilon}{1-\varepsilon}, \quad \tilde{f}(\{a, b\}) = 1 + \varepsilon.$$

The function f is nonnegative, monotone, and modular (hence submodular): $f(\{a, b\}) = f(\{a\}) + f(\{b\})$ holds because $1 + \frac{2\varepsilon}{1-\varepsilon} = \frac{1+\varepsilon}{1-\varepsilon}$. Value accuracy at level ε is checked set by set: on $\{a\}$, $\tilde{f} = (1 + \varepsilon)f$; on $\{b\}$, $\tilde{f} = f$; on $\{a, b\}$, $\tilde{f}(\{a, b\}) = 1 + \varepsilon = (1 - \varepsilon)\,\frac{1+\varepsilon}{1-\varepsilon} = (1 - \varepsilon)f(\{a, b\})$, the lower end of the band. At the pair $(S, e) = (\{a\}, b)$ the true gain is $d_b(\{a\}) = f(\{a, b\}) - f(\{a\}) = \frac{2\varepsilon}{1-\varepsilon} > 0$ while the predicted gain is $\tilde{d}_b(\{a\}) = \tilde{f}(\{a, b\}) - \tilde{f}(\{a\}) = 0$. A zero predicted gain at a positive true gain violates $\tilde{d}_e(S) \geq d_e(S)/\eta_u$ for every finite η_u, so no (η_u, η_o) of Definition 1 is finite for this $\tilde{f}$, and every bound of the form $L_K(\eta)$ is vacuous for it.

Part (ii). Let f be monotone submodular, not identically zero, and $\tilde{f} = (1 + M)f$ for $M > 0$. Every set S with $f(S) > 0$ witnesses $|\tilde{f}(S) - f(S)|/f(S) = M > \varepsilon$ for every $\varepsilon < M$, so $\tilde{f}$ is not value-accurate at any level below M. Predicted gains are $\tilde{d}_e(S) = (1 + M)\, d_e(S)$, a positive multiple of the true gains, so at every state the maximizers of $\tilde{d}_e(S)$ and of $d_e(S)$ are the same set of elements, and whichever of them tie breaking picks, the picked element has maximum true gain. Every step with positive chosen gain therefore contributes the factor $\max_{e \notin S^t} d_e(S^t)/d_{e_t}(S^t) = 1$ to Definition 3, giving $\eta^{\mathrm{sel}} = 1$, and Proposition 6 certifies the run at $L_K(1)$. (Gains are nonnegative by monotonicity, so no step has negative chosen gain.)

Part (iii). Fix $S \subseteq N$ and apply Definition 1 to the pair $(\emptyset, S)$ read through single elements: summing the telescoping of f and $\tilde{f}$ along any fixed enumeration of S, each increment of $\tilde{f}$ lies between $1/\eta_u$ and η_o times the corresponding increment of f, and increments of a monotone f are nonnegative, so

$$f(S)/\eta_u \;\leq\; \tilde{f}(S) \;\leq\; \eta_o\, f(S).$$

Subtracting $f(S)$ and dividing by $f(S) > 0$ (the case $f(S) = 0$ forces $\tilde{f}(S) = 0$ and contributes nothing) gives $\frac{1}{\eta_u} - 1 \leq \frac{\tilde{f}(S) - f(S)}{f(S)} \leq \eta_o - 1$, hence $|\tilde{f}(S) - f(S)|/f(S) \leq \max\{1 - 1/\eta_u,\ \eta_o - 1\}$.

No script checks the computations of this proof.

B.3 THE GUARANTEE (PROPOSITION 6)

This proof is included for completeness; the statement is essentially Theorem 1 of Goundan & Schulz (2007), restated in the prediction-error model of Section 2.

Let $S^0 = \emptyset, S^1, \ldots, S^K$ be the states of a run of predictive greedy with picks $e_0, \ldots, e_{K-1}$ and selection error η^{sel}, and put $r_t = f(O^*) - f(S^t)$.

Step 1: one greedy step covers a $1/(\eta^{\mathrm{sel}}K)$ share of the residual. Let $t < K$ be a step with $d_{e_t}(S^t) > 0$. Enumerate $O^* \setminus S^t = \{o_1, \ldots, o_m\}$, $m \leq |O^*| \leq K$, and telescope:

$$f(O^* \cup S^t) - f(S^t) = \sum_{i=1}^{m} \Big[f\big(S^t \cup \{o_1, \ldots, o_i\}\big) - f\big(S^t \cup \{o_1, \ldots, o_{i-1}\}\big) \Big] \;\leq\; \sum_{i=1}^{m} d_{o_i}(S^t),$$

one application of submodularity per term. Monotonicity gives $f(O^* \cup S^t) \geq f(O^*)$, so $r_t \leq \sum_{i=1}^{m} d_{o_i}(S^t) \leq m \max_{e \notin S^t} d_e(S^t) \leq K \max_{e \notin S^t} d_e(S^t)$. This is the only place where the count

$m \le K$ enters. At a step with $g_t = d_{e_t}(S^t) > 0$, Definition 3 gives $M_t = \max_{e \notin S^t} d_e(S^t) = a_t g_t \le \eta^{\mathrm{sel}}\, d_{e_t}(S^t)$ whenever η^{sel} is finite (Step 2 handles the zero steps), hence

$$d_{e_t}(S^t) \;\ge\; \frac{1}{\eta^{\mathrm{sel}} K}\, r_t, \qquad \text{and so} \qquad r_{t+1} = r_t - d_{e_t}(S^t) \;\le\; \Big(1 - \frac{1}{\eta^{\mathrm{sel}} K}\Big) r_t.$$

Step 2: the zero-gain steps. Monotonicity of f makes $g_t = d_{e_t}(S^t) \ge 0$ always, so the steps not covered by Step 1 are exactly those with $g_t = 0$, and Definition 3 distinguishes two cases. If the step is harmful ($M_t > 0$), then $a_t = \infty$, so $\eta^{\mathrm{sel}} = \infty$, $L_K(\infty) = 0$, and the claimed bound is the trivial one; nothing needs proving. If the step is benign ($M_t = 0$), the covering inequality $r_t \le K \max_{e \notin S^t} d_e(S^t) = K M_t$ of Step 1, whose derivation does not use positivity of the chosen gain, reads $r_t \le 0$: the run has already reached the optimal value, and $r_s = 0$ for every $s \ge t$ by monotonicity. Therefore, when η^{sel} is finite, the contraction of Step 1 may be applied at every step, treating the residual as 0 from the first zero step onwards.

Step 3: unrolling. With $r_0 = f(O^*)$, iterating K times gives $r_K \le (1 - \frac{1}{\eta^{\mathrm{sel}} K})^K f(O^*)$ and hence

$$f(S^K) = f(O^*) - r_K \;\ge\; \Big(1 - \big(1 - \tfrac{1}{\eta^{\mathrm{sel}} K}\big)^K\Big) f(O^*) \;=\; L_K(\eta^{\mathrm{sel}})\, f(O^*).$$

Since $\eta^{\mathrm{sel}} \ge 1$ and $K \ge 1$, the quantity $u = 1/(\eta^{\mathrm{sel}} K)$ lies in $(0, 1]$, and $1 - u \le e^{-u}$ gives $L_K(\eta^{\mathrm{sel}}) \ge 1 - e^{-1/\eta^{\mathrm{sel}}}$.

Step 4: the three error measures. At every state S^t of the run,

$$\max_{e \notin S^t} d_e(S^t) \;\le\; \eta_u \max_{e \notin S^t} \tilde{d}_e(S^t) \;=\; \eta_u\, \tilde{d}_{e_t}(S^t) \;\le\; \eta_u \eta_o\, d_{e_t}(S^t),$$

the first step by the lower band applied to the maximizing element, the middle equality by the greedy choice, the last by the upper band at (S^t, e_t). Reading the two ends as the step ratio a_t of Definition 3 gives, at every step with $g_t > 0$, $a_t \le \eta_u \eta_o$ when the band is the global one, and $a_t \le \eta^{\mathrm{tr}}$ when it is restricted to the states of the run; at a step with $g_t = 0$ the same display forces $M_t = 0$, so $a_t = 1$. Taking the maximum over the steps gives $\eta^{\mathrm{sel}} \le \eta^{\mathrm{tr}} \le \eta$ for every predictor with finite error ($\eta^{\mathrm{tr}} \le \eta$ because the states of the run are a subset of all sets); this is where the finiteness hypothesis of the last sentence of the Proposition is used. The map $x \mapsto 1 - \frac{1}{xK}$ is increasing on $x \ge 1$ with values in $[0, 1)$, so $x \mapsto (1 - \frac{1}{xK})^K$ is increasing and L_K is decreasing. Hence $L_K(\eta^{\mathrm{sel}}) \ge L_K(\eta^{\mathrm{tr}}) \ge L_K(\eta)$, and the bound of Step 3 holds a fortiori with η^{tr} or η in place of η^{sel}.

Remark 25 (stepwise product bound)**.** *When the per-step ratios are retained individually, the same unrolling gives*

$$f(S^K) \;\ge\; \Big(1 - \prod_{t=0}^{K-1} \big(1 - \tfrac{1}{K a_t}\big)\Big) f(O^*), \qquad 1/\infty = 0,$$

which degrades a harmful zero step to a contraction factor of 1 *instead of collapsing the whole bound to* 0*. For a run stopped after $K^* < K$ steps the product ranges over the executed steps only, and no L_K bound follows: the two-element instance of Section 2 caps the stopped run at ratio* $1/2$ *against* $L_2(1) = 3/4$*.*

No script certifies the steps of the main argument. That the bound is met with equality, and that the three rulers read $\eta^{\mathrm{sel}} = \eta^{\mathrm{tr}} = \hat{a} < \eta$ on the family of Appendix B.4 (also under the stepwise Definition 3: no zero step occurs in that family, see `results/K1_etasel_newdef_check.py`), is machine-checked by `results/F2_etasel_tight.py`.

B.4 PER-K TIGHTNESS (THEOREM 8)

Fix $K \ge 2$ and $\hat{a} > 1$, and put

$$a = 1 - \frac{1}{\hat{a} K} \in (0, 1), \qquad N = B \cup O, \quad |B| = |O| = K, \quad n = 2K.$$

For $S \subseteq N$ write $x = |S \cap B|$ and $y = |S \cap O|$, and define the pair $(f, \tilde{f})$ by

$$f(S) = F(x,y) = 1 - a^x\Big(1 - \frac{y}{K}\Big),$$
$$\tilde{f}(S) = G(x,y) = \begin{cases} 1 - a^x, & y = 0, \\ 1 - a^x + a^x\Big[(1-a) + \dfrac{(y-1)a}{K-1}\Big], & 1 \le y \le K. \end{cases} \tag{1}$$

Both are counting functions of (x, y), so Lemma 24 applies with $r = 2$, $N_1 = B$, $N_2 = O$. Note $F(0,0) = 0$ and $G(0,0) = 0$.

Step 1: a closed form for the second branch of G. For $1 \le y \le K$,

$$\begin{aligned} 1 - a^x + a^x(1-a) + \frac{(y-1)a^{x+1}}{K-1} &= 1 - a^{x+1} + \frac{(y-1)a^{x+1}}{K-1} \\ &= 1 - a^{x+1}\,\frac{(K-1)-(y-1)}{K-1} \;=\; 1 - a^{x+1}\,\frac{K-y}{K-1}. \end{aligned}$$

In particular $G(x,K) = 1$ for every x, and the two branches agree at $y = 1$: the second branch gives $1 - a^{x+1}$, which is $G(x,0) + a^x(1-a)$.

Step 2: the four gain identities. Writing Δ_x and Δ_y for the forward differences in x and y, the four cases are computed directly from equation 1 and Step 1:

$$\Delta_x F(x,y) = a^x(1-a)\,\frac{K-y}{K}, \qquad \Delta_y F(x,y) = \frac{a^x}{K}, \tag{2}$$
$$\Delta_x G(x,0) = a^x(1-a), \qquad \Delta_y G(x,0) = a^x(1-a), \tag{3}$$
$$\Delta_x G(x,y) = a^{x+1}(1-a)\,\frac{K-y}{K-1}, \qquad \Delta_y G(x,y) = \frac{a^{x+1}}{K-1}, \tag{4}$$

the last row for $1 \le y \le K$ in the x direction and for $1 \le y \le K-1$ in the y direction. The two entries of equation 3 are the same number because both read $(1 - a^{x+1}) - (1 - a^x)$: the first crosses no branch, and the second crosses from $G(x,0)$ to $G(x,1) = 1 - a^{x+1}$ by Step 1. Dividing equation 3 and equation 4 by the corresponding entries of equation 2 gives the four ratios, with $R := aK/(K-1)$:

$$\frac{\Delta_x G(x,0)}{\Delta_x F(x,0)} = 1, \qquad \frac{\Delta_y G(x,0)}{\Delta_y F(x,0)} = \frac{a^x(1-a)}{a^x/K} = K(1-a) = \frac{1}{\hat{a}},$$

and, in the range $1 \le y \le K-1$ where both are defined,

$$\frac{\Delta_x G(x,y)}{\Delta_x F(x,y)} = \frac{a^{x+1}(1-a)(K-y)/(K-1)}{a^x(1-a)(K-y)/K} = R, \qquad \frac{\Delta_y G(x,y)}{\Delta_y F(x,y)} = \frac{a^{x+1}/(K-1)}{a^x/K} = R,$$

where $K(1-a) = 1/\hat{a}$ is the definition of a. At $y = K$ the only remaining direction is x, and both gains vanish: $\Delta_x F(x,K) = 0$ by equation 2 and $\Delta_x G(x,K) = 0$ by Step 1. So a zero true gain always comes with a zero predicted gain, as Definition 1 demands.

Step 3: monotonicity and submodularity of f, case by case. The two first differences in equation 2 are nonnegative for $0 \le y \le K$ and $x \ge 0$, since $a \in (0,1)$. The four second differences are

$$\Delta_x\Delta_x F = -a^x(1-a)^2\,\frac{K-y}{K} \le 0, \qquad \Delta_y\Delta_x F = -\frac{a^x(1-a)}{K} \le 0,$$
$$\Delta_x\Delta_y F = \frac{a^{x+1} - a^x}{K} = -\frac{a^x(1-a)}{K} \le 0, \qquad \Delta_y\Delta_y F = \frac{a^x}{K} - \frac{a^x}{K} = 0 \le 0.$$

The two mixed differences coincide, as they must. Lemma 24 now gives that f is monotone and submodular on the whole of 2^N. (No claim is made about $\tilde{f}$; it is monotone but not submodular, which the model of Section 2 permits.)

Step 4: the global error of the pair. By Step 2 the set of single-element ratios $\tilde{d}_e(S)/d_e(S)$, over all S and all $e \notin S$ with $d_e(S) > 0$, is exactly $\{1,\ R,\ 1/\hat{a}\}$. Two sign computations order it:

$$R - 1 = \frac{aK - (K-1)}{K-1} = \frac{K - \frac{1}{\hat{a}} - K + 1}{K-1} = \frac{1 - 1/\hat{a}}{K-1} > 0, \qquad 1 - \frac{1}{\hat{a}} = \frac{\hat{a}-1}{\hat{a}} > 0,$$

both strict because $\hat{a} > 1$. Hence $\eta_o = \max\{1, R, 1/\hat{a}\} = R$ and $\eta_u = \max\{1, 1/R, \hat{a}\} = \hat{a}$, both attained, and

$$\eta = \eta_u \eta_o = \hat{a}\,\frac{aK}{K-1} = \frac{K(\hat{a} - \frac{1}{K})}{K-1} = \frac{K\hat{a}-1}{K-1}, \tag{5}$$

using $\hat{a}a = \hat{a} - 1/K$.

Step 5: the optimum. $1 - F(x,y) = a^x(K-y)/K \geq 0$, so $F \leq 1$ everywhere, and $F(0,K) = 1$. Since $|O| = K$, the value $f(O^*) = 1$ is attained at O, and $O^* = O$ may be taken.

Step 6: the greedy induction under adversarial ties. Claim: for $0 \leq t \leq K$ the state after t steps satisfies $S^t \subseteq B$ and $|S^t| = t$, that is $(x,y) = (t,0)$. For $t = 0$ this is $S^0 = \emptyset$. Assume it at $t < K$. At the state $(t,0)$ every candidate is either a remaining element of B, whose predicted gain is $\Delta_x G(t,0) = a^t(1-a)$, or an element of O, whose predicted gain is $\Delta_y G(t,0) = a^t(1-a)$; these are equal by equation 3, so every candidate is a maximizer and the step is a tie among all $2K - t$ of them. Adversarial tie breaking selects an element of B (which is possible because $t < K = |B|$), so S^{t+1} has $(x,y) = (t+1, 0)$. After K steps $S^K = B$ and

$$f(S^K) = F(K,0) = 1 - a^K = 1 - \Big(1 - \frac{1}{\hat{a}K}\Big)^K = L_K(\hat{a}),$$

which is $L_K(\hat{a}) f(O^*)$ by Step 5.

Step 7: the two run-dependent errors equal $\hat{a}$. At the state $S^t = (t,0)$ the true gains are $a^t(1-a) = a^t/(\hat{a}K)$ for an element of B and a^t/K for an element of O, by equation 2. Since $\hat{a} > 1$, the largest true gain at that state is a^t/K while the chosen element has gain $a^t/(\hat{a}K) > 0$, so every step has positive chosen gain, the stepwise a_t of Definition 3 reduce to the ratio form, and

$$a_t = \frac{\max_{e \notin S^t} d_e(S^t)}{d_{e_t}(S^t)} = \frac{a^t/K}{a^t/(\hat{a}K)} = \hat{a}$$

at every one of the K steps, so $\eta^{\mathrm{sel}} = \hat{a}$. states of the run matter, and at those states $y = 0$, so by Step 2 the only two ratios that occur are 1 (direction x) and $1/\hat{a}$ (direction y); therefore the trajectory-restricted factors are $\eta_o^{\mathrm{tr}} = \max\{1, 1/\hat{a}\} = 1$ and $\eta_u^{\mathrm{tr}} = \max\{1, \hat{a}\} = \hat{a}$, so $\eta^{\mathrm{tr}} = \hat{a}$ as well. Comparing with equation 5, $\eta - \hat{a} = \frac{K\hat{a}-1-(K-1)\hat{a}}{K-1} = \frac{\hat{a}-1}{K-1} > 0$, so on this family the selection and trajectory rulers agree and are strictly smaller than the global one.

Step 8: the same family read at fixed global error. Solving equation 5 for $\hat{a}$ gives $\hat{a} = (\eta(K-1)+1)/K$, hence $a = 1 - \frac{1}{\hat{a}K} = 1 - \frac{1}{\eta(K-1)+1}$ and

$$f(S^K) = 1 - a^K = 1 - \Big(1 - \frac{1}{\eta(K-1)+1}\Big)^K = U_K(\eta).$$

So the same instances that make Proposition 6 an equality at $\eta^{\mathrm{sel}} = \hat{a}$ realize only U_K when charged the global error η, which is the comparison recorded in Remark 15.

Steps 1 to 8 give Theorem 8: for every $K \geq 2$ and every $\hat{a} > 1$ the pair equation 1 on $2K$ elements has monotone submodular f, finite error, an adversarial-tie greedy run with $\eta^{\mathrm{sel}} = \eta^{\mathrm{tr}} = \hat{a}$, and output value exactly $L_K(\hat{a}) f(O^*)$.

The identities of this proof are machine-checked by `results/T5_symbolic.py` and `results/F2_etasel_tight.py`.

B.5 COHERENCE (LEMMA 9)

Let $S \subseteq N$ and $e, e' \notin S$ with $e \neq e'$, and assume $\tilde{d}_e(S) \geq \tilde{d}_{e'}(S)$.

The exchange identity. Expanding the same quantity $\tilde{f}(S \cup \{e, e'\}) - \tilde{f}(S)$ in the two possible orders, first e then e', and first e' then e:

$$\tilde{d}_e(S) + \tilde{d}_{e'}(S \cup \{e\}) \;=\; \tilde{f}(S \cup \{e, e'\}) - \tilde{f}(S) \;=\; \tilde{d}_{e'}(S) + \tilde{d}_e(S \cup \{e'\}).$$

This uses nothing but the definition of a set function. Subtracting and applying the hypothesis $\tilde{d}_e(S) - \tilde{d}_{e'}(S) \geq 0$,

$$\tilde{d}_{e'}(S \cup \{e\}) - \tilde{d}_e(S \cup \{e'\}) = \tilde{d}_{e'}(S) - \tilde{d}_e(S) \;\leq\; 0. \tag{6}$$

Part (i). Chain the two bands of Definition 1 around equation 6:

$$d_{e'}(S \cup \{e\}) \;\leq\; \eta_u \, \tilde{d}_{e'}(S \cup \{e\}) \;\leq\; \eta_u \, \tilde{d}_e(S \cup \{e'\}) \;\leq\; \eta_u \eta_o \, d_e(S \cup \{e'\}) \;=\; \eta \, d_e(S \cup \{e'\}),$$

the first inequality by the lower band at $(S \cup \{e\}, e')$, the second by equation 6, the third by the upper band at $(S \cup \{e'\}, e)$. Dividing by η gives (i).

Part (ii). The same two-order expansion applied to f reads $d_e(S) + d_{e'}(S \cup \{e\}) = d_{e'}(S) + d_e(S \cup \{e'\})$, that is

$$d_{e'}(S) - d_e(S) \;=\; d_{e'}(S \cup \{e\}) - d_e(S \cup \{e'\}).$$

Substituting part (i) in the form $-d_e(S \cup \{e'\}) \leq -\frac{1}{\eta} d_{e'}(S \cup \{e\})$,

$$d_{e'}(S) - d_e(S) \;\leq\; d_{e'}(S \cup \{e\}) - \frac{1}{\eta} \, d_{e'}(S \cup \{e\}) \;=\; \Big(1 - \frac{1}{\eta}\Big) d_{e'}(S \cup \{e\}),$$

which is (ii). Only $\eta \geq 1$ is used beyond the bands, and only to keep the factor $1 - \frac{1}{\eta}$ nonnegative when the inequality is applied later.

No script checks the computations of this proof. The two inequalities are used as constraint families in Appendix B.13, and the linear program they define is compared against the full-lattice program by `results/N3_K5_lattice.py`.

B.6 The exact worst case (Theorem 10)

Roadmap. The mechanism behind the value runs in the following order. The sharp form of Lemma 9, in Section 4.4, limits how much a single step can lose now and cost later at the same time; the two-phase trajectory of the attaining instances below is the shape that limit leaves available; the multipliers equation 8 turn that shape into a lower bound valid for every run, and the instances equation 11 attain it; Appendix B.7 reads the same certificate backwards and recovers the trajectory from the equality case; and the integer breakpoints appear there as the error levels at which the multipliers carrying that argument vanish. The coefficient bookkeeping of the certificate, the gain table of the instances, and the validity of the four constraint families (Appendix B.13) are kept apart from that line of reasoning and can be read afterwards.

Throughout this subsection $K \geq 2$, $\eta \geq 1$, $f(O^*) = 1$, and

$$k_1 = (K-1)\eta + 1, \qquad q = \frac{(K-1)\eta}{k_1} = 1 - \frac{1}{k_1}, \qquad V_j(\eta) = 1 - q^j \Big(1 - \frac{K-j}{K\eta}\Big).$$

Note $k_1 \geq K \geq 2$, so $q \in (0,1)$, and $(1-q)k_1 = 1$.

The reduced linear program. Every execution of predictive greedy induces $d_t = d_{e_t}(S^t)$ for $t = 0, \ldots, K-1$ and, after enumerating $O^* = \{o_1, \ldots, o_K\}$,

$$g_{t,i} = \begin{cases} d_{o_i}(S^t), & o_i \notin S^t, \\ 0, & o_i \in S^t, \end{cases} \qquad 0 \leq t \leq K, \; 1 \leq i \leq K.$$

Appendix B.13 derives, one family at a time and with the case $e_t \in O^*$ treated explicitly, that this vector satisfies

$$\begin{aligned} \textstyle\sum_i g_{t,i} + \sum_{s<t} d_s \;&\geq\; 1 \quad (\mathrm{sum}(t)), \\ d_t - g_{t,i}/\eta \;&\geq\; 0 \quad (\mathrm{pred}(t,i)), \\ g_{t,i} - g_{t+1,i} \;&\geq\; 0 \quad (\mathrm{mono}(t,i)), \\ \big(1 - \tfrac{1}{\eta}\big) g_{t+1,i} - g_{t,i} + d_t \;&\geq\; 0 \quad (\mathrm{cons}(t,i)), \end{aligned} \tag{7}$$

together with $d_t \geq 0$, $g_{t,i} \geq 0$, and that $\sum_{t<K} d_t = f(S^K)$ by telescoping. Consequently the minimum of $\sum_{t<K} d_t$ over the polyhedron equation 7 is at most the worst-case ratio $\rho_K(\eta)$: the polyhedron contains every realizable run and possibly more.

Lower bound: greedy is never worse than $\min_j V_j$. Fix a segment index j with $0 \leq j \leq K-1$ and let η lie in $[K-j, K-j+1]$ (in $[K,\infty)$ when $j=0$). Write $M = K\eta - (K-j)$ and define nonnegative numbers $\lambda_S(t)$, $\lambda_P(t)$, $\lambda_C(t)$ (one per constraint family, the same for every index i by the symmetry of equation 7 in i) as follows. For $j \geq 1$,

$$\begin{aligned} &\lambda_S(0) = \frac{q^{j-1}M}{Kk_1}, && \lambda_S(t) = \frac{q^{j-1-t}M}{k_1^2} \quad (1 \leq t \leq j-1), \\ &\lambda_S(j) = \frac{K-j+1-\eta}{k_1}, && \lambda_C(t) = \frac{q^{j-1-t}M}{Kk_1} \quad (t \leq j-1), \\ &\lambda_C(t) = \frac{K-1-t}{K(\eta-1)} \quad (t \geq j), && \lambda_P(t) = \frac{\eta-(K-t)}{K(\eta-1)} \quad (t \geq j), \end{aligned} \tag{8}$$

with $\lambda_S(t) = 0$ for $t > j$ and $\lambda_P(t) = 0$ for $t < j$; for $j = 0$,

$$\begin{aligned} &\lambda_S(0) = \frac{1}{\eta}, \qquad \lambda_S(t) = 0 \quad (t \geq 1), \\ &\lambda_C(t) = \frac{K-1-t}{K(\eta-1)}, \qquad \lambda_P(t) = \frac{\eta-(K-t)}{K(\eta-1)} \qquad (0 \leq t \leq K-1). \end{aligned} \tag{9}$$

All multipliers of the family mono are 0: the lower bound does not use that the optimal gains decrease along the run.

Nonnegativity. On the segment, $\eta \geq K-j \geq 1$ and $K\eta > K-j$, so $M > 0$; since also $q \in (0,1)$ and $k_1 > 0$, the three multipliers built from powers of q, namely $\lambda_S(0)$, $\lambda_S(t)$ for $1 \leq t \leq j-1$ and $\lambda_C(t)$ for $t \leq j-1$, are positive. For $\lambda_S(j)$, the segment condition $\eta \leq K-j+1$ is exactly $K-j+1-\eta \geq 0$, and equality holds at the right endpoint of the segment. For $\lambda_C(t)$ with $t \geq j$, $K-1-t \geq 0$ because $t \leq K-1$. For $\lambda_P(t)$ with $t \geq j$, $\eta-(K-t) \geq \eta-(K-j) \geq 0$ by the segment condition, with equality at $t=j$ at the left endpoint. The denominator $\eta-1$ is positive whenever $j \leq K-2$, since then $\eta \geq K-j \geq 2$. The one remaining case is $j = K-1$, whose segment is $[1,2]$: there the range $t \geq j$ contains only $t = K-1$, where the numerator $K-1-t$ vanishes identically and $\lambda_C(K-1)$ is defined to be 0, while $\lambda_P(K-1) = (\eta-1)/(K(\eta-1))$ is defined by cancellation to be $1/K$. So equation 8 has no singularity on any segment.

The weighted-sum identity. For an arbitrary vector $(d_t, g_{t,i})$, not necessarily feasible, consider

$$\begin{aligned} \Sigma \;:=\; &\sum_{t=0}^{j} \lambda_S(t)\Big[\sum_i g_{t,i} + \sum_{s<t} d_s - 1\Big] \;+\; \sum_{t=j}^{K-1}\sum_i \lambda_P(t)\Big[d_t - \frac{g_{t,i}}{\eta}\Big] \\ &+ \sum_{t=0}^{K-1}\sum_i \lambda_C(t)\Big[\big(1-\tfrac{1}{\eta}\big)g_{t+1,i} - g_{t,i} + d_t\Big]. \end{aligned} \tag{10}$$

The last summand of the third group is void, because $\lambda_C(K-1) = 0$ by its own formula in equation 8 and equation 9. Collecting coefficients in Σ gives three families of scalar identities. The coefficient of $g_{t,i}$, for $0 \leq t \leq K$, is $\lambda_S(t) - \lambda_P(t)/\eta - \lambda_C(t) + (1-\frac{1}{\eta})\lambda_C(t-1)$, with the conventions $\lambda_C(-1) = 0$, $\lambda_C(K) = 0$ and $\lambda_S(t) = \lambda_P(t) = 0$ outside the ranges listed above; the coefficient of d_t is $\sum_{s>t}\lambda_S(s) + K\big(\lambda_P(t) + \lambda_C(t)\big)$; the constant is $-\sum_{t\leq j}\lambda_S(t)$. The claim is that these equal 0, 1 and $-V_j$ respectively, so that $\Sigma = \sum_{t<K} d_t - V_j$ identically. Each is checked below; nothing is left to "direct expansion".

(a) The coefficient of $g_{t,i}$ vanishes. Take $j \geq 1$ first. For $t=0$ one has $\lambda_C(-1) = 0$ and $\lambda_P(0) = 0$ (as $0 < j$), so the coefficient is $\lambda_S(0) - \lambda_C(0) = 0$, the two entries of equation 8 being literally the same number. For $1 \leq t \leq j-1$, again $\lambda_P(t) = 0$, and dividing the coefficient by the positive quantity $q^{j-1-t}M/(Kk_1)$ turns it into

$$\frac{K}{k_1} - 1 + \Big(1-\frac{1}{\eta}\Big)q = \frac{K-k_1}{k_1} + \frac{\eta-1}{\eta}\cdot\frac{(K-1)\eta}{k_1} = \frac{(K-1)(1-\eta)+(\eta-1)(K-1)}{k_1} = 0,$$

using $K - k_1 = K - (K-1)\eta - 1 = (K-1)(1-\eta)$. For $t = j$, write $u = K - j$, so that $\eta \in [u, u+1]$ and $M = K\eta - u$, and multiply the coefficient by $K\eta k_1(\eta - 1) > 0$:

$$\underbrace{K\eta(\eta-1)(u+1-\eta)}_{\text{from } \lambda_S(j)} \underbrace{-\,k_1(\eta-u)}_{\text{from } \lambda_P(j)/\eta} \underbrace{-\,(u-1)\eta k_1}_{\text{from } \lambda_C(j)} \underbrace{+\,(\eta-1)^2(K\eta-u)}_{\text{from } (1-1/\eta)\lambda_C(j-1)}.$$

The two middle terms combine as $-k_1\big[(\eta-u)+(u-1)\eta\big] = -k_1 u(\eta-1)$, so with $A = \eta - 1$ the whole expression is $A\big[K\eta(u+1-\eta) + A(K\eta-u) - k_1 u\big]$, and the bracket expands to

$$K\eta u + K\eta - K\eta^2 + K\eta^2 - K\eta - \eta u + u - k_1 u = u\big[K\eta - \eta + 1 - k_1\big] = u\,[\,k_1 - k_1\,] = 0.$$

For $j+1 \le t \le K-1$ one has $\lambda_S(t) = 0$, and multiplying by $K\eta(\eta-1)$ the coefficient becomes

$$-(\eta-K+t) - \eta(K-1-t) + (\eta-1)(K-t) = -\eta + K - t - \eta K + \eta + \eta t + \eta K - \eta t - K + t = 0.$$

For $t = K$ only the last term survives and $\lambda_C(K-1) = 0$. The multiplication by $\eta - 1$ in the last two computations is legitimate on every segment with $j \le K-2$, where $\eta \ge 2$; on the segment $j = K-1$, which is $[1,2]$, both sides of each identity are rational functions of η that are continuous at $\eta = 1$ after the cancellation recorded above, so the identity at $\eta = 1$ is the limit of the identity on $(1,2]$. For $j = 0$ the first computation is replaced by the single case $t = 0$: multiplying by $K\eta(\eta-1)$, $K(\eta-1) - (\eta-K) - \eta(K-1) = K\eta - K - \eta + K - \eta K + \eta = 0$, and the cases $1 \le t \le K-1$ and $t = K$ are the ones just done.

(b) The coefficient of d_t equals 1. For $t \ge j$ the sum $\sum_{s>t} \lambda_S(s)$ is empty, and

$$K\big(\lambda_P(t) + \lambda_C(t)\big) = K \cdot \frac{(\eta-K+t)+(K-1-t)}{K(\eta-1)} = \frac{\eta-1}{\eta-1} = 1,$$

which at $j = K-1$, $\eta = 1$ reads $K(\frac{1}{K}+0) = 1$ with the cancelled values. For $t = j-1$ (so $j \ge 1$), $\lambda_P(j-1) = 0$ and the only surviving λ_S is $\lambda_S(j)$, so the coefficient is

$$\frac{K-j+1-\eta}{k_1} + K \cdot \frac{M}{Kk_1} = \frac{(K-j+1-\eta)+(K\eta-K+j)}{k_1} = \frac{(K-1)\eta+1}{k_1} = 1.$$

For $t \le j-2$ the value is unchanged from $t+1$ to t: subtracting the two coefficients and using $\lambda_P(t) = \lambda_P(t+1) = 0$,

$$\begin{aligned}\lambda_S(t+1) + K\big[\lambda_C(t) - \lambda_C(t+1)\big] &= \frac{q^{j-2-t}M}{k_1^2} + K \cdot \frac{Mq^{j-2-t}}{Kk_1}\,(q-1) \\ &= \frac{q^{j-2-t}M}{k_1^2} - \frac{Mq^{j-2-t}}{k_1^2} = 0,\end{aligned}$$

because $q - 1 = -1/k_1$. Downward induction from $t = j-1$ therefore gives the value 1 for every $t \le j-1$ as well.

(c) The constant equals $-V_j$. For $j \ge 1$, the middle multipliers form a geometric sum with ratio q, which is summed with $1 - q = 1/k_1$:

$$\sum_{t=1}^{j-1} \lambda_S(t) = \frac{M}{k_1^2} \sum_{m=0}^{j-2} q^m = \frac{M}{k_1^2} \cdot \frac{1-q^{j-1}}{1-q} = \frac{M\big(1-q^{j-1}\big)}{k_1},$$

so that, using $M + (K-j) = K\eta$ and hence $M + (K-j+1-\eta) = K\eta - \eta + 1 = k_1$,

$$\sum_{t=0}^{j} \lambda_S(t) = \frac{1}{k_1}\Big[\frac{Mq^{j-1}}{K} + M - Mq^{j-1} + (K-j+1-\eta)\Big] = \frac{1}{k_1}\Big[k_1 - Mq^{j-1}\frac{K-1}{K}\Big].$$

The definition of q gives $\frac{K-1}{K} = \frac{qk_1}{K\eta}$, so $Mq^{j-1}\frac{K-1}{K} = \frac{Mq^j k_1}{K\eta}$ and

$$\sum_{t=0}^{j} \lambda_S(t) = 1 - \frac{Mq^j}{K\eta} = 1 - q^j\,\frac{K\eta - (K-j)}{K\eta} = 1 - q^j\Big(1 - \frac{K-j}{K\eta}\Big) = V_j.$$

For $j = 0$ the sum is the single term $\lambda_S(0) = 1/\eta$, and $V_0 = 1 - (1 - \frac{K}{K\eta}) = 1/\eta$.

Conclusion of the lower bound. By (a), (b), (c), identity equation 10 reads $\sum_{t<K} d_t - V_j = \Sigma$ for every vector $(d_t, g_{t,i})$. At a point feasible for equation 7 every bracket in Σ is nonnegative and every multiplier is nonnegative, so $\Sigma \geq 0$ and $\sum_{t<K} d_t \geq V_j$. Since every run of predictive greedy at error η gives such a feasible point with objective $f(S^K)/f(O^*)$,

$$\rho_K(\eta) \;\geq\; V_j(\eta) \qquad \text{on the segment of } j.$$

Which V_j is smallest. For $0 \leq i \leq K-2$,

$$\begin{aligned} V_i - V_{i+1} &= q^i\Big[q - \frac{q(K-i-1)}{K\eta} - 1 + \frac{K-i}{K\eta}\Big] \\ &= q^i\Big[-\frac{1}{k_1} + \frac{1}{K\eta}\Big(1 + \frac{K-i-1}{k_1}\Big)\Big] \;=\; \frac{q^i\,(K-i-\eta)}{K\eta k_1}, \end{aligned}$$

where the middle step uses $q - 1 = -1/k_1$ and $(K-i) - q(K-i-1) = 1 + \frac{K-i-1}{k_1}$, and the last step uses $k_1 - K\eta = 1 - \eta$. On the segment $\eta \in [K-j, K-j+1]$ the sign of $K - i - \eta$ is nonnegative for $i \leq j-1$ and nonpositive for $i \geq j$, so the sequence $V_0, \ldots, V_{K-1}$ decreases up to index j and increases after it: $\min_i V_i = V_j$ there, and neighbouring segments agree at the integer endpoints. The breakpoints are therefore the integers $2, \ldots, K$, and $\rho_K(\eta) \geq \min_i V_i(\eta)$ for every $\eta \geq 1$. At $\eta \geq K$ the active index is $j = 0$ and $\min_i V_i = V_0 = 1/\eta$. The closed forms displayed in Theorem 10 for $K \in \{2,3,4\}$ are the specializations of $\min_j V_j$; for instance at $K = 3$ and $j = 2$, $V_2 = 1 - \frac{4\eta^2}{(2\eta+1)^2} \cdot \frac{3\eta-1}{3\eta} = \frac{16\eta+3}{3(2\eta+1)^2}$.

The identities of this proof are machine-checked by `results/N1_dual_certificate.py`, and the $K \in \{2,3,4\}$ closed forms by `results/T3_K3_closed_form.py`.

Upper bound: the value V_j is attained. Fix $K \geq 2$, $0 \leq j \leq K-1$, and η with $\eta \geq K-j$, and fix any split $\eta_u\eta_o = \eta$ with $\eta_u, \eta_o \geq 1$ (the scripts use $\eta_u = \eta_o = \sqrt{\eta}$). Partition a ground set of $n = 2K$ elements into

$$C\ (|C| = j), \qquad P\ (|P| = K-j), \qquad O\ (|O| = K),$$

write $x = |S \cap C|$, $z = |S \cap P|$, $y = |S \cap O|$, and put

$$\delta_j = \frac{q^j}{K\eta}, \qquad \tilde{\delta}_j = \eta_o\delta_j = \frac{q^j}{K\eta_u}, \qquad W(0) = \frac{k_1}{K\eta_u}, \qquad W(y) = \frac{(K-y)\eta_o}{K}\ \ (1 \leq y \leq K),$$

and $\chi(y) = \mathbf{1}[y < K]$. The instance is

$$f(S) = 1 - q^x\Big(1 - \frac{y}{K}\Big) + z\,\delta_j\,\chi(y), \qquad \tilde{f}(S) = W(0) - q^x W(y) + z\,\eta_o\delta_j\,\chi(y). \tag{11}$$

Both are counting functions of (x, z, y), so Lemma 24 applies with $r = 3$. Two normalizations are immediate: $f(\emptyset) = 1 - 1 = 0$ and $\tilde{f}(\emptyset) = W(0) - W(0) = 0$.

Marginal gains. Directly from equation 11, for $y < K$,

$$\Delta_C f = q^x(1-q)\frac{K-y}{K}, \qquad \Delta_P f = \delta_j, \qquad \Delta_O f = \begin{cases} q^x/K, & y \leq K-2, \\ q^x/K - z\delta_j, & y = K-1, \end{cases}$$

and at $y = K$ all three gains are 0. On the predictor side, $\Delta_C \tilde{f} = q^x(1-q)W(y)$ and $\Delta_P \tilde{f} = \eta_o\delta_j$, while

$$\Delta_O \tilde{f} = \begin{cases} q^x\big(W(0) - \frac{(K-1)\eta_o}{K}\big), & y = 0, \\ q^x\eta_o/K, & 1 \leq y \leq K-2, \\ q^x\eta_o/K - z\eta_o\delta_j, & y = K-1. \end{cases}$$

Two identities are used repeatedly: $(1-q)k_1 = 1$ and

$$W(0) - \frac{(K-1)\eta_o}{K} = \frac{(K-1)\eta + 1}{K\eta_u} - \frac{(K-1)\eta_u\eta_o}{K\eta_u} = \frac{1}{K\eta_u}. \tag{12}$$

Monotonicity and submodularity of f. The three first differences are nonnegative: $\Delta_C f \geq 0$ and $\Delta_P f > 0$ are immediate from $q \in (0,1)$, and at $y = K-1$

$$\Delta_O f = \frac{q^x}{K} - z\,\frac{q^j}{K\eta} \;\geq\; 0 \iff \eta\, q^{x-j} \;\geq\; z,$$

which holds because $x \leq j$ makes $q^{x-j} \geq 1$ and $z \leq K-j \leq \eta$ by the standing assumption $\eta \geq K-j$. This is the only place where the segment condition is consumed, and it is where the breakpoints come from. The nine second differences are: $\Delta_C\Delta_C f = -q^x(1-q)^2\frac{K-y}{K} \leq 0$; $\Delta_O\Delta_C f = -q^x(1-q)/K \leq 0$ for every $y \leq K-1$ (at $y = K-1$ the gain drops from $q^x(1-q)/K$ to 0, the same decrement); $\Delta_C\Delta_O f = -q^x(1-q)/K \leq 0$; $\Delta_P\Delta_C f = \Delta_C\Delta_P f = \Delta_P\Delta_P f = 0$; $\Delta_O\Delta_P f = 0$ for $y \leq K-2$ and $-\delta_j \leq 0$ at $y = K-1$; $\Delta_P\Delta_O f = 0$ for $y \leq K-2$ and $-\delta_j \leq 0$ at $y = K-1$; $\Delta_O\Delta_O f = 0$ for $y \leq K-3$ and $-z\delta_j \leq 0$ at $y = K-2$. Lemma 24 then gives that f is monotone and submodular.

The error is exactly (η_u, η_o). Divide each predicted gain by the corresponding true gain, using equation 12 in the case $y = 0$ of direction O. Every single-element ratio $\tilde{d}/d$ takes one of three values:

$$\frac{\Delta_C \tilde{f}}{\Delta_C f}\Big|_{y=0} = W(0) = \frac{k_1}{K\eta_u}, \qquad \frac{\Delta_O \tilde{f}}{\Delta_O f}\Big|_{y=0} = \frac{1}{\eta_u}, \qquad \eta_o \text{ in all remaining cases,}$$

the remaining cases being direction C with $y \geq 1$, direction P, and direction O with $y \geq 1$. The three are ordered by $k_1 - K = (K-1)(\eta-1) \geq 0$ and $K\eta - k_1 = \eta - 1 \geq 0$, that is

$$\frac{1}{\eta_u} \;\leq\; \frac{k_1}{K\eta_u} \;\leq\; \frac{K\eta}{K\eta_u} = \eta_o.$$

The outer two are attained, so $\max \tilde{d}/d = \eta_o$ and $\max d/\tilde{d} = \eta_u$: the pair sits exactly on the band of Definition 1, and its error is not merely bounded by η.

The optimum. $f(O) = f(0,0,K) = 1$, and for every S with $|S| \leq K$ one has $f(S) \leq 1$: at $y = K$ the value is exactly 1, and at $y \leq K-1$, $f(S) = 1 - q^x\frac{K-y}{K} + z\delta_j \;\leq 1$ because $z\delta_j \leq (K-j)\frac{q^j}{K\eta} \leq \frac{q^x}{K} \leq q^x\frac{K-y}{K}$ whenever $y \leq K-1$, using $z \leq K-j \leq \eta$ and $q^j \leq q^x$. So $f(O^*) = 1$ with $O^* = O$.

The per-step gain table. The run is the following; the table lists, at each state, the true and predicted gains of the three kinds of candidate.

step t	state (x,z,y)	candidate	true gain	predicted gain	picked
$t < j$	$(t,0,0)$	$e \in C$	q^t/k_1	$q^t/(K\eta_u)$	yes
		$e \in P$	$q^j/(K\eta)$	$q^j/(K\eta_u)$	no
		$e \in O$	q^t/K	$q^t/(K\eta_u)$	no (tie)
$t \geq j$	$(j, t-j, 0)$	$e \in P$	$q^j/(K\eta)$	$q^j/(K\eta_u)$	yes
		$e \in O$	q^j/K	$q^j/(K\eta_u)$	no (tie)

The entries follow from the gain formulas above: at $y = 0$ a C-step has true gain $q^t(1-q) = q^t/k_1$ and predicted gain $q^t(1-q)W(0) = q^tW(0)/k_1 = q^t/(K\eta_u)$; a P-step has predicted gain $\eta_o\delta_j = q^j/(K\eta_u)$; an O-step at $y = 0$ has predicted gain $q^x/(K\eta_u)$ by equation 12.

The greedy induction. Claim: for $0 \leq t \leq K$ the state after t steps is $(x,z,y) = (\min(t,j),\ \max(t-j,0),\ 0)$. It holds at $t = 0$. Assume it at $t < j$. By the table the maximum predicted gain at that state is $q^t/(K\eta_u)$, attained by every remaining element of C and every element of O, while every element of P scores $q^j/(K\eta_u)$, which is strictly smaller because $q \in (0,1)$ and $j > t$. Adversarial tie breaking picks an element of C, so the state becomes $(t+1, 0, 0)$. Now assume the claim at t with $j \leq t < K$. The state is $(j, t-j, 0)$, C is exhausted, and the predicted gains of the remaining P elements and of the O elements are both $q^j/(K\eta_u)$: the step is a tie and the adversary picks an element of P, which is available because $t - j < K - j = |P|$. After K steps $S^K = C \cup P$ and

$$f(S^K) = 1 - q^j + (K-j)\,\delta_j = 1 - q^j + \frac{(K-j)q^j}{K\eta} = 1 - q^j\Big(1 - \frac{K-j}{K\eta}\Big) = V_j(\eta),$$

which is $V_j(\eta)f(O^*)$ because $f(O^*) = 1$. Choosing j with $\eta \in [K-j, K-j+1]$ (that is j the minimizing index of the previous paragraph) gives $\rho_K(\eta) \le \min_i V_i(\eta)$, which together with the lower bound is Theorem 10.

The role of the ties. Every one of the K steps above is a tie with an element of O, so adversarial tie breaking is load bearing. On the steps $t \ge j$ the tie is forced by the band: the prediction constraint of equation 7 holds with equality at the optimum, so $\eta_o\, d_P = d_O/\eta_u$, and the largest value the band allows a P element and the smallest it allows an O element coincide. Under strict preference the same value is an infimum rather than a maximum: for η in the interior of the segment $[K-j, K-j+1]$, replacing η by $\eta' < \eta$ inside f while declaring the band at η makes all preferences strict and gives ratio $V_j(\eta') \to V_j(\eta)$ as $\eta' \uparrow \eta$.

The identities of this proof are machine-checked by `results/N2_check.py`.

B.7 Structure of the attaining runs (Proposition 12)

Throughout this subsection the notation is that of Appendix B.6, $f(O^*) = 1$, and η lies in the interior of a segment: either $K - j < \eta < K - j + 1$ for some j with $1 \le j \le K-1$, or $j = 0$ and $\eta > K$. Write $\gamma = 1 - \frac{1}{\eta}$ and $r_t = 1 - \sum_{s<t} d_s$, so that $r_0 = 1$ and $r_{t+1} = r_t - d_t$. Fix a run of predictive greedy with adversarial ties whose value attains the exact worst case, $\sum_{t<K} d_t = V_j(\eta)$; Appendix B.13 makes the vector $(d_t, g_{t,i})$ it induces feasible for equation 7. Nothing below assumes that the run avoids O^* or that the numbers $g_{t,i}$ are equal across i; both come out of the argument.

Step 1: which multipliers are positive. On the interior of the segment, $M = K\eta - (K-j) > 0$, $q \in (0,1)$, $k_1 > 0$, and $\eta > 1$: for $j \le K-2$ the segment gives $\eta > K - j \ge 2$, and for $j = K-1$ the segment is $(1,2)$. Reading the signs off equation 8 and equation 9:

- $\lambda_S(t) > 0$ exactly for $0 \le t \le j$. The entries with $t \le j-1$ are positive multiples of M, and $\lambda_S(j) = (K-j+1-\eta)/k_1 > 0$ because $\eta < K-j+1$ strictly inside the segment, while $\lambda_S(t) = 0$ for $t > j$ by definition. For $j = 0$ the single entry is $\lambda_S(0) = 1/\eta > 0$.
- $\lambda_C(t) > 0$ exactly for $0 \le t \le K-2$. For $t \le j-1$ the entry is a positive multiple of M; for $j \le t \le K-2$ it is $(K-1-t)/(K(\eta-1)) > 0$; and $\lambda_C(K-1) = 0$.
- $\lambda_P(t) > 0$ exactly for $j \le t \le K-1$. It is 0 for $t < j$ by definition, and for $t \ge j$ its numerator is $\eta - (K-t) \ge \eta - (K-j) > 0$, again strictly inside the segment.
- every multiplier of the family mono is 0.

At an integer η one of the two strict inequalities above becomes an equality, which is the content of Remark 13: $\lambda_P(j)$ vanishes at the left endpoint of the segment and $\lambda_S(j)$ at the right endpoint.

Step 2: every supported constraint holds with equality. Identity equation 10 reads $\sum_{t<K} d_t - V_j = \Sigma$ for every vector, and Σ is a sum of terms $\lambda_r\,\mathrm{slack}_r$ with $\lambda_r \ge 0$ throughout and $\mathrm{slack}_r \ge 0$ at a feasible point. The run attains the value, so the left-hand side is 0, so every term vanishes, so

$$\mathrm{sum}(t)\ (0 \le t \le j), \qquad \mathrm{cons}(t,i)\ (0 \le t \le K-2), \qquad \mathrm{pred}(t,i)\ (j \le t \le K-1)$$

hold with equality, for every index i. The constraints carrying a zero multiplier, namely $\mathrm{sum}(t)$ for $t > j$, $\mathrm{cons}(K-1,i)$ and every $\mathrm{mono}(t,i)$, stay valid but are not forced; Step 7 uses two of them as inequalities.

Step 3: the geometric phase, $t < j$. Let $t < j$; the range is empty when $j = 0$. Both $\mathrm{sum}(t)$ and $\mathrm{sum}(t+1)$ carry indices at most j, so $\sum_i g_{t,i} = r_t$ and $\sum_i g_{t+1,i} = r_{t+1}$. Summing the K equalities $\mathrm{cons}(t,i)$ over i, which is legitimate because $t \le j-1 \le K-2$,

$$\gamma \sum_i g_{t+1,i} - \sum_i g_{t,i} + K d_t = 0, \qquad \text{that is} \qquad \gamma\, r_{t+1} = r_t - K d_t.$$

Substituting $r_{t+1} = r_t - d_t$ gives $d_t(K-\gamma) = r_t(1-\gamma) = r_t/\eta$, and $K - \gamma = K - 1 + \frac{1}{\eta} = k_1/\eta$, so

$$d_t = \frac{r_t}{k_1}, \qquad r_{t+1} = r_t\Big(1 - \frac{1}{k_1}\Big) = q\, r_t.$$

With $r_0 = 1$ this gives $r_t = q^{\,t}$ and $d_t = q^{\,t}/k_1$ for every $t < j$, and $r_j = q^{\,j}$.

Step 4: the switch at $t = j$. The index j lies in the support of both λ_S and λ_P, so $\mathrm{pred}(j,i)$ holds with equality for every i, which is $g_{j,i} = \eta\, d_j$, and $\mathrm{sum}(j)$ holds with equality, which is $\sum_i g_{j,i} = r_j = q^{\,j}$ (for $j = 0$ this is $r_0 = 1 = q^0$). Hence $K\eta\, d_j = q^{\,j}$, that is

$$d_j = \frac{q^{\,j}}{K\eta}, \qquad g_{j,i} = \frac{q^{\,j}}{K} \quad \text{for every } i.$$

The K numbers $g_{j,i}$ come out equal; this is a conclusion of the equality case, not a symmetry imposed on the program, which treats the indices i separately.

Step 5: the frozen phase, $j \le t \le K-2$. Both $\mathrm{pred}(t,i)$ and $\mathrm{pred}(t+1,i)$ hold with equality, so $g_{t,i} = \eta d_t$ and $g_{t+1,i} = \eta d_{t+1}$, and $\mathrm{cons}(t,i)$ holds with equality, so

$$\gamma\, g_{t+1,i} = g_{t,i} - d_t = (\eta - 1)d_t, \qquad \text{while} \qquad \gamma\, g_{t+1,i} = \gamma\eta\, d_{t+1} = (\eta-1)d_{t+1}.$$

Since $\eta > 1$ this forces $d_{t+1} = d_t$ and then $g_{t+1,i} = g_{t,i}$. With Step 4, $d_t = q^{\,j}/(K\eta)$ and $g_{t,i} = q^{\,j}/K$ for every t with $j \le t \le K-1$ and every i.

Step 6: the optimal marginals before the switch. For $t < j$ the equality $\mathrm{cons}(t,i)$ reads $g_{t,i} = d_t + \gamma\, g_{t+1,i}$. Going downwards from $g_{j,i} = q^{\,j}/K$ and assuming $g_{t+1,i} = q^{\,t+1}/K$,

$$g_{t,i} = \frac{q^{\,t}}{k_1} + \gamma\,\frac{q^{\,t+1}}{K} = \frac{q^{\,t}}{K}\Big[\frac{K}{k_1} + \frac{\eta-1}{\eta}\cdot\frac{(K-1)\eta}{k_1}\Big] = \frac{q^{\,t}}{K}\cdot\frac{K + (K-1)(\eta-1)}{k_1} = \frac{q^{\,t}}{K},$$

because $K + (K-1)(\eta-1) = (K-1)\eta + 1 = k_1$. Downward induction gives $g_{t,i} = q^{\,t}/K$ for every $t \le j$ and every i, one index i at a time.

Step 7: the last state. The multiplier $\lambda_C(K-1)$ is 0, so Step 2 does not force $\mathrm{cons}(K-1,i)$; the constraint is valid all the same (Lemma 30), and so is $\mathrm{mono}(K-1,i)$ (Lemma 29). Write $d = d_{K-1}$, $g = g_{K-1,i}$ and $h = g_{K,i}$. Rearranged as the sharp form of Section 4.4, the two constraints read

$$d - \frac{g}{\eta} \;\ge\; \Big(1 - \frac{1}{\eta}\Big)(g-h) \qquad \text{and} \qquad g - h \;\ge\; 0.$$

Step 2 forces $\mathrm{pred}(K-1,i)$ with equality, that is $d = g/\eta$, so the left-hand side of the first display is 0; since $\eta > 1$ the two displays leave $h = g = q^{\,j}/K$. Every variable of the run is therefore determined, and the values are those of Proposition 12. As a check, $\sum_{t<K} d_t = (1-q^{\,j}) + (K-j)q^{\,j}/(K\eta) = V_j(\eta)$, and the gain table of Appendix B.6 shows that the instances equation 11 follow exactly this trajectory, so the forced trajectory is realized.

What the argument delivers and what it leaves open. Every $g_{K,i}$ equals $q^{\,j}/K > 0$, while the zeroing convention of Appendix B.13 sets $g_{K,i} = 0$ whenever $o_i \in S^K$. So no optimal element is selected: an attaining run in this range of η is disjoint from the fixed O^*. The reduction of Appendix B.13 still has to admit runs that select optimal elements, and the equality analysis excludes them only in the narrower situation treated here. The conclusion is about the vector $(d_t, g_{t,i})$ and about nothing else: the pair $(f, \tilde f)$ is not determined by it, and neither are the true marginals of the candidates the run does not select. In particular the largest true marginal at a state, the quantity that Definition 3 compares the chosen gain against, is not pinned by this argument. At an integer η the argument stops at Step 1, and Remark 13 records that the conclusion itself fails there.

Two consequences of the same certificate are recorded in Remark 14. For the first, the slack of $\mathrm{pred}(t,i)$ decomposes in case (b) of Appendix B.13, with $P = \tilde d_{e_t}(S^t)$ and $Q = \tilde d_{o_i}(S^t)$, as

$$d_t - \frac{g_{t,i}}{\eta} = \underbrace{\frac{\eta_o d_t - P}{\eta_o}}_{\text{upper band at } (S^t, e_t)} + \underbrace{\frac{P - Q}{\eta_o}}_{\text{greedy}} + \underbrace{\frac{\eta_u Q - g_{t,i}}{\eta_u \eta_o}}_{\text{lower band at } (S^t, o_i)},$$

the three-term counterpart of the decomposition of the $\mathrm{cons}(t,i)$ slack in Lemma 30. Steps 2 and 7 put both slacks at zero, the coherence one for $t \le K-2$ and the greedy-choice one at $t = K-1$, and each term of each decomposition is nonnegative, so $P = Q$ at every step: the selected element and every optimal element carry the same predicted gain, and the run is a sequence of ties. For the second, if the run outputs $V_j(\eta) + \varepsilon$ instead of $V_j(\eta)$, then equation 10 gives $\sum_r \lambda_r \,\mathrm{slack}_r = \varepsilon$ with all terms nonnegative, hence $\mathrm{slack}_r \le \varepsilon/\lambda_r$ for every constraint with $\lambda_r > 0$. That bound is stated for one run at a time; the multipliers themselves degenerate at the segment endpoints, and no uniform statement over η and K is derived from it here.

The recurrences of Steps 3 to 6, the rearrangement of Step 7 and the two $K = 3$, $\eta = 2$ sequences of Remark 13 are machine-checked by `results/H_J3_gate_check.py`. The same script also fixes the objective of equation 7 at the exact rational $V_j(\eta)$ and minimizes and maximizes each coordinate separately, for $K = 2, \ldots, 6$, every j and two error levels per segment: on all 40 optimal faces the 2,480 linear programs collapse every coordinate range to the values above, with a largest deviation of about $4 \cdot 10^{-15}$, and no symmetry constraint is added to the program. What no script checks is the assembly of Steps 1 to 7 into a statement about a run of arbitrary size: the face enumeration is finite, and the induction above carries the general case.

B.8 THE CEILING (THEOREM 16)

Fix $\eta_u, \eta_o \ge 1$, $K \ge 1$, $n \ge 2K$, and a constant $c > 0$.

The construction. Let $\tilde{f}(S) = c|S|$, which does not depend on the hidden optimum, and for a K-set O with $B = N \setminus O$ let

$$f_O(S) = c\Big(\frac{|S \cap B|}{\eta_o} + \eta_u |S \cap O|\Big).$$

f_O is modular with positive coefficients, hence monotone and submodular with $f_O(\emptyset) = 0$. Its single-element gains are c/η_o on B and $c\eta_u$ on O, against the constant predicted gain c, so the ratios $\tilde{d}/d$ take the two values η_o (on B) and $1/\eta_u$ (on O), and the error is exactly (η_u, η_o) with both factors attained. Because both functions are modular, the gain of any pair (A, B') is the sum of single-element gains along any chain from A to $A \cup B'$, so every all-pairs ratio is a weighted average of the two single-element values and the all-pairs error is exactly (η_u, η_o) as well. Since $\eta_u \ge 1 \ge 1/\eta_o$, the K largest coefficients are those of O and $f_O(O^*) = f_O(O) = c\eta_u K$.

Deterministic algorithms. The transcript of a deterministic algorithm querying $\tilde{f}$ is the same for every O, so its output T, $|T| \le K$, is fixed before O is chosen. With $n \ge 2K$ there is a K-set $O \subseteq N \setminus T$, and for it

$$\frac{f_O(T)}{f_O(O^*)} = \frac{c|T|/\eta_o}{c\eta_u K} \le \frac{cK/\eta_o}{c\eta_u K} = \frac{1}{\eta_u\eta_o} = \frac{1}{\eta}.$$

Randomized algorithms. Let O be uniform among K-subsets, independent of the random string. The output T is again independent of O, and writing $m = |T| \le K$, $\mathbb{E}|T \cap O| = mK/n$ and $\mathbb{E}|T \cap B| = m(1 - K/n)$, so

$$\mathbb{E}\Big[\frac{f_O(T)}{f_O(O^*)}\Big] = \frac{m(1-\frac{K}{n})/\eta_o + \eta_u mK/n}{\eta_u K} = \frac{m}{K}\Big[\frac{1-\frac{K}{n}}{\eta_u\eta_o} + \frac{K}{n}\Big] \le \Big(1 - \frac{K}{n}\Big)\frac{1}{\eta} + \frac{K}{n},$$

and $\min_O \le \mathbb{E}_O$ turns this into a statement about one instance.

The matching upper bound by exhaustive search. For any S, enumerate $S = \{s_1, \ldots, s_m\}$ and telescope both functions along the same chain $S_0 = \emptyset \subset S_1 \subset \cdots \subset S_m = S$. The upper band applied to each of the m terms gives $\tilde{f}(S) = \sum_i \tilde{d}_{s_i}(S_{i-1}) \le \eta_o \sum_i d_{s_i}(S_{i-1}) = \eta_o f(S)$, and the lower band gives $\tilde{f}(S) \ge f(S)/\eta_u$. Let $\hat{S}$ maximize $\tilde{f}$ over all sets of size at most K. Then

$$f(\hat{S}) \;\ge\; \frac{\tilde{f}(\hat{S})}{\eta_o} \;\ge\; \frac{\tilde{f}(O^*)}{\eta_o} \;\ge\; \frac{f(O^*)}{\eta_u\eta_o} = \frac{f(O^*)}{\eta},$$

the middle inequality by the choice of $\hat{S}$. So the constant $1/\eta$ is reached by an algorithm that is unrestricted in queries but makes no use of structure, which is the sense in which the ceiling is met.

No script checks the computations of this proof.

B.9 ASYMPTOTICS (COROLLARY 18)

Fix $\eta \geq 1$. For L_K, the substitution $u = 1/(\eta K) \in (0, 1]$ and the expansion $\ln(1-u) = -u - u^2/2 - \dots$ give $K \ln\bigl(1 - \frac{1}{\eta K}\bigr) = -\frac{1}{\eta} + O(1/K)$, so $L_K(\eta) \to 1 - e^{-1/\eta}$; moreover $(1 - \frac{x}{K})^K$ increases in K for $x \in (0, 1]$, so the convergence of $L_K(\eta)$ is monotone from above. For $\rho_K = \min_j V_j$, fix η and let K grow. The minimizing index is the unique j with $\eta \in [K-j, K-j+1]$, that is $j = K - \lfloor \eta \rfloor$ for noninteger η and either endpoint value at an integer η, so $K - j$ stays bounded while $j \to \infty$. With $k_1 = (K-1)\eta + 1$,

$$\ln q^{\,j} = j \ln\Bigl(1 - \frac{1}{k_1}\Bigr) = -\frac{j}{k_1} + O\Bigl(\frac{j}{k_1^2}\Bigr), \qquad \frac{j}{k_1} = \frac{K - \lfloor \eta \rfloor}{(K-1)\eta + 1} \longrightarrow \frac{1}{\eta}, \qquad \frac{j}{k_1^2} \longrightarrow 0,$$

so $q^{\,j} \to e^{-1/\eta}$, while the second factor of V_j satisfies $1 - \frac{K-j}{K\eta} = 1 - \frac{\lfloor \eta \rfloor}{K\eta} \to 1$. Hence $\rho_K(\eta) = V_j(\eta) \to 1 - e^{-1/\eta}$, the same limit as L_K, and by Remark 15 the two curves differ by $O(1/K)$ at fixed η.

The limit identities of this proof are machine-checked, in the form used by Theorem 19, by `results/N5_asymptotics.py`.

B.10 BOUNDED-QUERY HARDNESS (THEOREM 19)

In this subsection ”information-theoretic” is used with the following meaning, stated here at its first use: the bound depends only on the number and the size of the queries that the algorithm makes and uses no computational assumption, so nothing below is a statement about running time.

The family. Fix integers $1 \leq \tau < K < n$ and a real $\theta \geq 1$, and write $a = a_\theta$, $A = A_\theta$, $B = B_\theta$ for the constants of Section 4.8. For a hidden K-set $O \subseteq N$ and $S \subseteq N$ put $x = |S \setminus O|$ and $y = |S \cap O|$, and define

$$F_O(S) = \theta^{-1/2}\Bigl[1 - a^x\Bigl(1 - \frac{y}{K}\Bigr)\Bigr], \qquad G_O(S) = \begin{cases} 1 - a^{\,x+y}, & y \leq \tau, \\ 1 - a^{\,x+\tau}\,\dfrac{K-y}{K-\tau}, & y \geq \tau, \end{cases}$$

together with the balanced-region profile $\hat{G}(s) = 1 - a^s$. The two branches of G_O agree at $y = \tau$. Both functions depend on S only through the pair (x, y), so Lemma 24 applies to them with the partition $N = (N \setminus O) \cup O$. The instance is $f = F_O$ and $\tilde{f} = G_O$; the prefactor $\theta^{-1/2}$ places the symmetric split and cancels from every ratio below.

Every edge of the count grid. On an edge with positive true gain put $r = \Delta G_O/(\sqrt{\theta}\,\Delta F_O)$, the predicted-to-true gain ratio of that edge. Subtracting the two displays gives the following exhaustive table.

Direction	Starting state	r
x	$0 \leq y \leq \tau$	$a^y K/(K-y)$
x	$\tau \leq y < K$	A
y	$0 \leq y \leq \tau - 1$	a^y/θ
y	$\tau \leq y < K$	A

An x-edge starting at $y = K$ has zero true and zero predicted gain, and no y-edge starts at $y = K$, so neither carries a band constraint. The edge from $y = \tau - 1$ to $y = \tau$ is in the third row and the edge from $y = \tau$ to $y = \tau + 1$ is in the fourth, so the junction of the two branches needs no separate extension argument, and the edges leaving the balanced strip are already in rows two and four. Every x cancels, so the table is the same for all $n > K$.

Write $h_y = a^y K/(K-y)$ for the first row. Then

$$\frac{h_{y+1}}{h_y} - 1 = \frac{(\theta - 1)K + y}{\theta K(K - y - 1)} \;\geq\; 0,$$

so $1 = h_0 \le h_y \le h_\tau = A$ for $0 \le y \le \tau$. The third row a^y/θ decreases from $1/\theta$ to $a^{\tau-1}/\theta = 1/(\theta B)$. Since $A \ge 1$ and $B \ge 1$, the largest and the smallest values of r over the whole grid are A and $1/(\theta B)$, and both are attained on actual edges because $n > K$ and $1 \le \tau < K$. In the notation of Definition 1, where the band reads $1/\eta_u \le \tilde{d}_e(S)/d_e(S) \le \eta_o$ and $\tilde{d}_e(S)/d_e(S) = \sqrt{\theta}\, r$, the smallest admissible factors of the unscaled pair are therefore

$$\eta_o^{\mathrm{act}} = \sqrt{\theta}\, A, \qquad \eta_u^{\mathrm{act}} = \sqrt{\theta}\, B, \qquad \eta^{\mathrm{act}} = \theta AB = \frac{\theta K - 1}{K - \tau} = \Psi(\theta),$$

and both factors are at least 1, as Definition 1 requires.

A prescribed split of the error. Replacing $\tilde{f} = G_O$ by βG_O with $\beta = \eta_o/(\sqrt{\theta}A)$ multiplies every r by β, so the largest predicted-to-true ratio becomes η_o and the smallest becomes $1/\eta_u$, with $\eta_u\eta_o = \eta^{\mathrm{act}}$ unchanged. The canonical profile $\hat{G}$ is multiplied by the same constant and remains a function of $|S|$ alone, so the rescaling carries no information about the identity of O. The choice $\beta = \sqrt{B/A}$ returns the symmetric split $\eta_u = \eta_o = \sqrt{\eta^{\mathrm{act}}}$.

The true objective and its optimum. Write $\bar{F} = \sqrt{\theta}\, F_O$, a function of (x, y). Its first differences are

$$\Delta_x \bar{F} = a^x(1-a)\Big(1 - \frac{y}{K}\Big), \qquad \Delta_y \bar{F} = \frac{a^x}{K},$$

nonnegative on every feasible edge, and its second differences are

$$\Delta_x^2 \bar{F} = -a^x(1-a)^2\Big(1 - \frac{y}{K}\Big), \qquad \Delta_y\Delta_x \bar{F} = \Delta_x\Delta_y \bar{F} = -\frac{a^x(1-a)}{K}, \qquad \Delta_y^2 \bar{F} = 0,$$

all nonpositive. Lemma 24 then makes F_O a normalized monotone submodular function on every finite ground set. Also $0 \le \bar{F} \le 1$ and $\bar{F}(O) = 1$, so $\max_{|S|\le K} F_O(S) = F_O(O) = \theta^{-1/2}$ and the optimal set is $O^* = O$. If $|T| \le K$ and $T \cap O = \emptyset$, then

$$\frac{F_O(T)}{F_O(O^*)} = 1 - a^{|T|} \;\le\; 1 - a^K = L_K(\theta).$$

Counting. For a fixed S with $|S| \le K$ and a uniformly random K-subset O of the n-element ground set, the event $\{|S \cap O| > \tau\}$ is the union, over the $(\tau+1)$-subsets $R \subseteq S$, of the events $\{R \subseteq O\}$, and $\Pr[R \subseteq O] = \prod_{i=0}^{\tau} \frac{K-i}{n-i} \le (K/n)^{\tau+1}$, since $(K-i)n \le K(n-i)$ for $n \ge K$. A union bound over at most $\binom{K}{\tau+1} \le K^{\tau+1}/(\tau+1)!$ sets R gives

$$\Pr\big[|S \cap O| > \tau\big] \;\le\; \binom{K}{\tau+1}\Big(\frac{K}{n}\Big)^{\tau+1} \;\le\; \frac{K^{2\tau+2}}{(\tau+1)!\, n^{\tau+1}}.$$

The integrality of τ enters at this step, which is the reason for $\tau = \lceil c \rceil + 1$ rather than $\tau = c + 1$ at real c.

The canonical transcript, and the deterministic statement. Let $\mathcal{A}$ be deterministic, make at most $Q = n^c$ queries of size at most K, and return a set of size at most K. Run it against the canonical oracle $S \mapsto \hat{G}(|S|)$, rescaled by β if a split was prescribed, and let $S_1, \ldots, S_Q$ and T_0 be the queries and the output it produces. These are fixed sets, determined before O is drawn. Summing the counting bound over the Q canonical queries and adding $\Pr[T_0 \cap O \ne \emptyset] \le \mathbb{E}|T_0 \cap O| = |T_0|K/n \le K^2/n$,

$$\Pr\big[\exists i:\ |S_i \cap O| > \tau \ \text{ or } \ T_0 \cap O \ne \emptyset\big] \;\le\; \frac{K^{2\tau+2}}{(\tau+1)!\, n^{\tau+1-c}} + \frac{K^2}{n}.$$

Under $n \ge 4K^{c+2}$ the first term is at most

$$\frac{K^{\,c(c+1-\tau)}}{(\tau+1)!\, 4^{\,\tau+1-c}} \;\le\; \frac{1}{16\,(\tau+1)!} \;\le\; \frac{1}{32},$$

because $\tau \ge c+1$ makes the exponent of K nonpositive and $\tau + 1 - c \ge 2$; the second term is $K^2/n \le 1/(4K^c) \le 1/4$. The sum is at most $9/32$, so at least one K-set O makes every canonical query balanced and misses T_0; fix such an O. Induction on i: if the first $i-1$ answers of G_O agree with the canonical ones, determinism makes the ith query equal to S_i, and $|S_i \cap O| \le \tau$ puts S_i in the first branch, where $x + y = |S_i|$ and $G_O(S_i) = \hat{G}(|S_i|)$. So all Q answers agree, the actual output is T_0, and $T_0 \cap O = \emptyset$ gives $f(T_0)/f(O^*) \le 1 - a^K = L_K(\theta)$.

The order of the two arguments matters: the union bound is taken over the canonical transcript, which does not depend on O. A union bound over the queries an algorithm actually issues against G_O would not by itself make those queries independent of O.

Randomized algorithms, by averaging. Fix the random seed r; then $\mathcal{R}_r$ is deterministic, with canonical queries and canonical output $T_0^{(r)}$, and let E_r be the event that all of its canonical queries are balanced. On E_r the run against G_O returns $T_0^{(r)}$. Every output has at most K elements, so its counts satisfy $x \le K$, and $a \in (0,1)$ gives

$$1 - a^x\Big(1 - \frac{y}{K}\Big) \;\le\; 1 - a^K + \frac{y}{K},$$

while off E_r the output ratio is at most 1. Hence, pointwise in O,

$$\frac{f_O(T)}{f_O(O^*)} \;\le\; 1 - a^K + \frac{|T_0^{(r)} \cap O|}{K} + \mathbf{1}[E_r^c].$$

Averaging over a uniform O bounds the middle term by K/n and the last one by the query union bound above; averaging over r, exchanging the two finite expectations and using $\min_O \le \mathbb{E}_O$ gives a single O with the additive ε_n of Theorem 19. Only the averaging direction of the minimax principle is used; no minimax theorem is invoked.

Calibration, comparison with the symmetric band, and the limit. Choosing $\theta = \bar{\theta} = \Psi^{-1}(\eta) = (\eta(K-\tau)+1)/K$ makes the pair's error exactly η and turns the value bound into

$$L_K(\bar{\theta}) = 1 - \Big(1 - \frac{1}{\bar{\theta}K}\Big)^K = 1 - \Big(1 - \frac{1}{\eta(K-\tau)+1}\Big)^K = H_{K,\tau}(\eta),$$

the constant of Theorem 19. Since $\eta\tau \ge 1$ gives $\bar{\theta} \le \eta$ and L_K is decreasing, $H_{K,\tau}(\eta) \ge L_K(\eta)$, so Theorem 19 and Proposition 6 do not cross at any finite K.

For the same integer τ, $\Psi(\theta) = \theta AB \le \theta \max\{A,B\}^2 = \Phi(\theta)$, with equality only at $A = B$. Hence $\bar{\theta} \ge \hat{\eta} = \Phi^{-1}(\eta)$ wherever the latter is defined, and $H_{K,\tau}(\eta) = L_K(\bar{\theta}) \le L_K(\hat{\eta})$: the constant above is not weaker than the one the symmetric-band calibration produces. The two differ already at $(K,\tau,\theta) = (4,1,4)$, where the pair has factors $(\eta_u,\eta_o) = (2,\frac{5}{2})$ and error 5, while $\Phi(4) = \frac{25}{4}$; a family calibrated through Φ^{-1} therefore has error at most, and in general below, the prescribed level, which is what the earlier statement of "error exactly η" did not account for.

The inflation relative to θ is $\delta(\theta) = AB - 1 = (\tau - 1/\theta)/(K-\tau)$, so $K\delta(\theta) \to \tau - 1/\theta$ with τ and θ fixed. With c and η fixed, $K/(\eta(K-\tau)+1) \to 1/\eta$ and $H_{K,\tau}(\eta) \to 1 - e^{-1/\eta}$. No branch has to be inverted and no case distinction enters the design parameter.

The edge table, the extremal factors, the exact product Ψ, the second differences, the calibration identity and both limits are re-checked by `results/J2_core_oracles.py`, symbolically for general parameters and on an exact rational count grid; the extremes and the calibration comparison are recomputed independently by `results/H3_j2_recheck.py`. The limit check of `results/N5_asymptotics.py`, cited at the end of Appendix B.9, is the corresponding statement for $L_K(\hat{\eta})$ and so belongs to the superseded calibration; the limit of $H_{K,\tau}$ is the one checked in `results/J2_core_oracles.py`. No script certifies the counting bound, the transcript induction or the two averagings: those are the hand-written part of this subsection, and finite computations over instances do not range over adaptive algorithms.

B.11 The greedy budget (Corollary 22)

Fix $K \ge 2$, $\eta > 1$ and $n \ge 4K^5$, and take the family of Appendix B.10 with $\tau = 1$. The constraint $\eta \ge (K-1)/(K-\tau)$ of Theorem 19 reads $\eta \ge 1$ and holds vacuously, and $\bar{\theta} - 1 = (\eta-1)(K-1)/K \ge 0$, so the calibration is admissible and the value bound of the family is

$L_K(\bar{\theta}) = H_{K,1}(\eta) = U_K(\eta)$. The edge table, the exact error $\Psi(\theta)$, the prescribed-split rescaling and the optimum $O^* = O$ are exactly as in Appendix B.10; nothing in them refers to the query budget. Only the counting changes.

Counting at the budget $Q = nK$. At $\tau = 1$ a query S, $|S| \le K$, is unbalanced when $|S \cap O| \ge 2$, which is the union over the pairs $R \subseteq S$ of the events $\{R \subseteq O\}$. For a uniformly random K-subset O,

$$\Pr[R \subseteq O] = \frac{K(K-1)}{n(n-1)} = \Big(\frac{K}{n}\Big)^2 - \frac{K(n-K)}{n^2(n-1)} \;\le\; \Big(\frac{K}{n}\Big)^2,$$

so one query is unbalanced with probability at most $\binom{K}{2}(K/n)^2$, and the canonical transcript's $Q = nK$ queries $S_1, \ldots, S_Q$ together with its output T_0 fail with probability at most

$$nK\binom{K}{2}\Big(\frac{K}{n}\Big)^2 + \Pr[T_0 \cap O \ne \emptyset] \;\le\; \frac{K^4(K-1)}{2n} + \frac{K^2}{n} \;\le\; \frac{K^5}{2n} + \frac{K^2}{n}.$$

Under $n \ge 4K^5$ the first term is at most $\frac{1}{8}$ and the second is at most $1/(4K^3) \le \frac{1}{32}$, so the total is at most $\frac{5}{32} < \frac{1}{2}$: some K-set O makes every canonical query balanced and misses T_0. The transcript induction of Appendix B.10, with balanced now meaning $|S_i \cap O| \le 1$, then forces the actual run to reproduce the canonical one and gives $f(T_0)/f(O^*) \le 1 - a^K = L_K(\bar{\theta}) = U_K(\eta)$, with error exactly η and any prescribed split realized by the β-rescaling. The ceiling $1/\eta$ holds for every deterministic algorithm at $n \ge 2K$ (Theorem 16), and $4K^5 \ge 2K$, so the two caps combine into $\min\{U_K(\eta), 1/\eta\}$ for the worst-case ratio.

The only property of the budget that the computation uses is $Q \le n^2/(4K^4)$, which $Q = nK$ satisfies exactly when $n \ge 4K^5$. By contrast, covering the budget nK on the scale of Theorem 19 needs $n^c \ge nK$, hence $c > 1$ and $\tau = \lceil c \rceil + 1 \ge 3$, and delivers only the weaker constant $H_{K,3}(\eta) > H_{K,1}(\eta)$; running the count at the exact budget is what keeps the $\tau = 1$ constant.

Randomized algorithms. The averaging of Appendix B.10, applied verbatim with the union bound above, yields for every randomized algorithm of the class a single instance with $\mathbb{E}[f(T)/f(O^*)] \le U_K(\eta) + K/n + K^5/(2n)$; the corollary states the slightly larger $\varepsilon_n = K^2/n + K^5/(2n)$, which also matches the deterministic output count. The quantifier order is that of Theorem 19: the instance may depend on the algorithm, and the expectation is over the algorithm's randomness only.

The width of the interval. For the gap statement of Remark 23, expanding both closed forms at fixed η in $1/K$ (with $m = \lfloor \eta \rfloor$ and $\rho_K = V_{K-m}$ on the active branch) gives

$$U_K(\eta) - \rho_K(\eta) = \frac{e^{-1/\eta}\, m\,(2\eta - m - 1)}{2\eta^2} \cdot \frac{1}{K^2} + O\Big(\frac{1}{K^3}\Big):$$

the constant and the $1/K$ coefficients of the two series agree (the latter is the $c(\eta)$ of Appendix B.9), and the $1/K^2$ coefficients differ by $c'(\eta)$ as displayed. At integer η the branches $m = \eta$ and $m = \eta - 1$ of c' agree, with common value $e^{-1/\eta}(\eta-1)/(2\eta)$, so c' is continuous although the $1/K^2$ coefficient of ρ_K alone is piecewise. Both series, the cancellation at order $1/K$, the closed form of c' and its positivity for $\eta > 1$ are checked in `results/L1_table.py` (section 2, symbolically; section 3 numerically in exact rationals up to $K = 400$, with one Richardson step); the two ends of the interval are tabulated in `results/L1_table.csv`. All of this is conditional on Theorem 10 through the closed form of ρ_K.

B.12 Queries of arbitrary size: a construction direction with finite evidence

Theorem 19 restricts every query to at most K elements. A companion construction removes that restriction and pays for it in the budget. For $K \ge 3$ and $\eta > 1$ put $k_1 = \eta(K-1)+1$, $q = 1 - 1/k_1$, $\nu = \eta/(\eta-1)$, $j = \min\{K, \max\{0, K+1-\lceil \eta \rceil\}\}$, $m^* = \lceil \eta K \rceil - 1$ and $t^* = j + m^*$, and let

$$D = \max\Big\{\frac{q^j}{K\eta},\; \frac{q^j\left(\nu^{m^*}/K - 1\right)}{\eta(\nu^{m^*} - 1) - m^*}\Big\}, \qquad W_K(\eta) = 1 - q^j + (K-j)D.$$

The family is the one of `results/F3_hardness_full.tex`: a hidden K-set O, a true objective that is piecewise geometric in $x = |S \setminus O|$ and flattens at $x > t^*$, and a predictor that agrees with an O-free profile $\hat{G}_{|S|}$ whenever $|S \cap O| \le 1$ and also whenever $|S \setminus O| > t^*$. The second half of that condition is what removes the query-size restriction: a query can break the simulation only if it meets O twice and has at most $t^* + K$ elements, so large queries need no concentration argument at all. Two caveats are recorded with the draft. If the tolerance is raised to $|S \cap O| \le \tau$ with $\tau \ge 2$, the family is infeasible at every inflation, by an explicit two-constraint certificate at all 48 tested configurations. If instead the predictor is required to be a function of $|S|$ on the whole two-sided band $\big||S \cap O| - K|S|/n\big| \le \tau$, the family is feasible exactly when $\tau = 1$ and $n > K(t^* + 1)$, and infeasible at every inflation otherwise; which of the two formalizations a reader accepts is the open judgement call of that draft.

Candidate bound (not a theorem of this paper). The intended statement is: for $K \ge 3$, $\eta > 1$ and $n \ge 4K(t^* + K)$ (which forces $n \ge 8K^2$), every deterministic algorithm that makes at most $Q \le n^2/\big(2K^2(t^* + K)^2\big)$ queries to $\tilde{f}$, of arbitrary size, and returns a set of size at most K, meets a K-set O for which the instance $(f, \tilde{f})$ has monotone submodular f, single-element error exactly η, $O^* = O$, and output $T_{\mathcal{A}}$ with

$$\frac{f(T_{\mathcal{A}})}{f(O^*)} \;\le\; W_K(\eta) \;=\; V_j(\eta) + (K - j)\Big(D - \frac{q^j}{K\eta}\Big),$$

and, for randomized algorithms, the same bound in expectation up to an additive $K/n + QK^2(t^* + K)^2/(2n^2)$. It is a candidate because the admissibility of the family at general parameters is exactly what the finite evidence below does not yet establish.

Since $t^* + K \le (2+\eta)K + 1$, the admissible budget is $Q = \Omega\big(n^2/((2+\eta)^2 K^4)\big)$: for fixed K and η this is a polynomial budget of degree approaching 2, and not every polynomial; a uniform reading such as $n^{2-o(1)}$ would additionally require restricting how K and η grow with n, which is not done here. Within the 48 checked configurations, the exponent 2 is a property of this family rather than of the analysis: it is the exponent in $\Pr[|S \cap O| \ge 2] \le (Ks/n)^2/2$ for a query of size s, and raising it would require the O-oblivious window to tolerate $|S \cap O| \le \tau$ with $\tau \ge 2$, which is the first of the two regimes recorded above as infeasible; no claim is made beyond those configurations or for extensions of the family. The constant is the reduced-LP segment value V_j of Theorem 10 plus an excess that decays quickly in K: at $\eta = 2$ it is $3.2 \cdot 10^{-4}$, $5.9 \cdot 10^{-7}$, $4.5 \cdot 10^{-12}$ and $5.2 \cdot 10^{-22}$ at $K = 4, 8, 16, 32$, and $L_K(\eta) \le V_j(\eta) \le W_K(\eta) < U_K(\eta)$ with $W_K(\eta) \to 1 - e^{-1/\eta}$.

What this supplementary statement does not carry is a general-K argument. The closed forms of j and m^* have numerical support at 264 parameter points and no proof; monotonicity, submodularity and the normalization of the family are enumerated exactly at 42 (K, η) points and have no general-K symbolic derivation; the concentration lemma, the simulation induction, the assembly and the averaging are hand arguments that no script checks, and the randomized version additionally uses an inequality on the family that no script has checked. The full draft, with its per-step declarations, is `results/F3_hardness_full.tex` in the supplementary material; the enumerations behind it are `results/N4_check.py`, `results/F3_check.py` and `results/N4_symbolic.py`.

B.13 Validity of the reduced constraints

This subsection supplies the step consumed by Appendix B.6: every constraint of equation 7 is a valid inequality, so the linear program is a relaxation of the set of realizable runs and its optimum is a lower bound on the worst-case ratio for every K.

Let f be monotone submodular with $f(\emptyset) = 0$, let $\tilde{f}$ be an arbitrary set function with $\tilde{f}(\emptyset) = 0$ obeying the band of Definition 1, and let $e_0, \ldots, e_{K-1}$ and $S^0 = \emptyset, \ldots, S^K$ be the picks and states of a run of predictive greedy, so that $e_t \in \arg\max_{e \notin S^t} \tilde{d}_e(S^t)$. Fix an optimal set $O^* = \{o_1, \ldots, o_K\}$ with $f(O^*) = 1$ (if $|O^*| < K$, set $g_{t,i} = 0$ for the unused indices; every case (a) below covers them). Put $d_t = d_{e_t}(S^t)$ and

$$g_{t,i} = \begin{cases} d_{o_i}(S^t), & o_i \notin S^t, \\ 0, & o_i \in S^t, \end{cases}$$

the second line being the *zeroing convention*, which is the device that handles a step at which greedy selects an element of O^*: from that step on the corresponding variable is 0 rather than undefined. It is consistent with reading $g_{t,i}$ as a marginal gain, since an element already in S^t contributes nothing further. Finally $r_t = 1 - \sum_{s<t} d_s$, which telescopes to $r_t = f(O^*) - f(S^t)$.

Throughout, $t \in \{0, \dots, K-1\}$, $i \in \{1, \dots, K\}$, and the three cases are (a) $o_i \in S^t$; (b) $o_i \notin S^t$ and $o_i \neq e_t$; (c) $o_i = e_t$, which is the case $e_t \in O^*$.

Lemma 26 (signs). *$d_t \geq 0$ and $g_{t,i} \geq 0$ for all t, i.*

Proof. Each of these numbers is a marginal gain of f or, under the zeroing convention, equal to 0. This step uses monotonicity of f. □

Lemma 27 (coverage, $\mathrm{sum}(t)$). *$\sum_{i=1}^{K} g_{t,i} \geq r_t$ for every t.*

Proof. Let $I_t = \{i : o_i \notin S^t\}$, so that $\sum_i g_{t,i} = \sum_{i \in I_t} d_{o_i}(S^t)$ by the zeroing convention; this is exactly where a step with $e_s \in O^*$ for some $s < t$ is absorbed, the indices of the already picked optimal elements dropping out of the sum. Order $I_t = \{i_1, \dots, i_m\}$ and telescope

$$f(S^t \cup O^*) - f(S^t) = \sum_{\ell=1}^{m} \Big[f\big(S^t \cup \{o_{i_1}, \dots, o_{i_\ell}\}\big) - f\big(S^t \cup \{o_{i_1}, \dots, o_{i_{\ell-1}}\}\big) \Big] \leq \sum_{\ell=1}^{m} d_{o_{i_\ell}}(S^t),$$

one application of submodularity per term. Monotonicity gives $f(S^t \cup O^*) \geq f(O^*)$, so $\sum_i g_{t,i} \geq f(O^*) - f(S^t) = r_t$. □

Lemma 28 (greedy choice inside the band, $\mathrm{pred}(t,i)$). *$d_t \geq g_{t,i}/\eta$ for every t, i.*

Proof. Case (a): $g_{t,i} = 0$ and $d_t \geq 0$ by Lemma 26. Case (b): chain

$$d_t = d_{e_t}(S^t) \;\geq\; \frac{\tilde{d}_{e_t}(S^t)}{\eta_o} \;\geq\; \frac{\tilde{d}_{o_i}(S^t)}{\eta_o} \;\geq\; \frac{d_{o_i}(S^t)}{\eta_u \eta_o} = \frac{g_{t,i}}{\eta},$$

the first inequality by the upper band at (S^t, e_t), the second by the greedy choice ($o_i \notin S^t$ makes o_i eligible at step t), the third by the lower band at (S^t, o_i). Case (c): here $g_{t,i} = d_{o_i}(S^t) = d_{e_t}(S^t) = d_t$ and the claim reads $d_t \geq d_t/\eta$, which holds because $d_t \geq 0$ and $\eta \geq 1$. No band inequality is needed in this case, so the constraint does not degrade when greedy happens to pick an optimal element. □

Lemma 29 (single-element submodularity, $\mathrm{mono}(t,i)$). *$g_{t+1,i} \leq g_{t,i}$ for every t, i.*

Proof. Case (a): $o_i \in S^t \subseteq S^{t+1}$, so both sides are 0. Case (b): $o_i \notin S^{t+1}$ as well, so $g_{t+1,i} = d_{o_i}(S^{t+1}) \leq d_{o_i}(S^t) = g_{t,i}$ by submodularity in its diminishing-returns form. Case (c): $o_i = e_t \in S^{t+1}$, so $g_{t+1,i} = 0$ while $g_{t,i} = d_t \geq 0$ by monotonicity. This is the only place among the four families where the case $e_t \in O^*$ replaces a submodularity argument by a monotonicity argument. □

Lemma 30 (coherence, $\mathrm{cons}(t,i)$). *$(1 - \frac{1}{\eta}) g_{t+1,i} \geq g_{t,i} - d_t$ for every t, i.*

Proof. Case (a): the left side is 0 and the right side is $-d_t \leq 0$. Case (b): abbreviate $d = d_t$, $g = g_{t,i}$, $h = g_{t+1,i}$ and, for the predicted gains, $P = \tilde{d}_{e_t}(S^t)$, $Q = \tilde{d}_{o_i}(S^t)$, $R = \tilde{d}_{o_i}(S^{t+1})$. Because the four values of a set function on S^t, $S^{t+1} = S^t \cup \{e_t\}$, $S^t \cup \{o_i\}$ and $S^t \cup \{e_t, o_i\}$ determine both orders of insertion, the gains of e_t at $S^t \cup \{o_i\}$ are $d + h - g$ (true) and $P + R - Q$ (predicted). The claim is then the sum of three nonnegative terms:

$$\Big(1 - \frac{1}{\eta}\Big) h - g + d \;=\; \underbrace{\frac{\eta_o (d + h - g) - (P + R - Q)}{\eta_o}}_{\text{upper band at } (S^t \cup \{o_i\},\, e_t)} + \underbrace{\frac{P - Q}{\eta_o}}_{\text{greedy}} + \underbrace{\frac{\eta_u R - h}{\eta_u \eta_o}}_{\text{lower band at } (S^{t+1},\, o_i)},$$

an identity in the six symbols; the first brace is nonnegative by Definition 1 at the off-trajectory state $S^t \cup \{o_i\}$, the second because o_i is eligible at step t and e_t was chosen, the third by the lower band. (Equivalently one may route the claim through part (ii) of Lemma 9; the display above is the same argument with its certificate written out.) Case (c): $g_{t+1,i} = 0$ by the zeroing convention and $g_{t,i} - d_t = d_{e_t}(S^t) - d_{e_t}(S^t) = 0$, so the constraint holds with equality. Lemma 9, which requires $e \neq e'$, is not invoked in this case, which is what closes the gap this subsection exists to close. □

Lemma 31 (relaxation)**.** *For every instance $(f, \tilde{f})$ satisfying Definition 1 with $\eta = \eta_u \eta_o$, every optimal set O^* and every run of predictive greedy, the vector $(d_t, g_{t,i})$ is feasible for equation 7 and its objective value equals $f(S^K)/f(O^*)$. Hence the optimum of equation 7 is a lower bound on the worst-case ratio of predictive greedy at budget K and error η.*

Proof. Feasibility is Lemmas 26 to 30. The objective is $\sum_{t<K} d_t = f(S^K)$ by telescoping, and $f(O^*) = 1$ by the normalization. Every realizable run gives a feasible point, so the minimum over the feasible set is at most the minimum over realizable runs. □

Remark 32 (what is and is not claimed)**.** *Lemma 31 gives one direction only. The reverse direction, that every feasible point of equation 7 is realized by some instance, is not claimed here; the attaining instances of Appendix B.6 supply what the theorem needs instead, and the agreement of equation 7 with the full-lattice program has been checked for $K \leq 5$ and at nineteen (K, η) test points.*

Remark 33 (which error measure each family tolerates)**.** *The families $\mathrm{sum}(t)$ and $\mathrm{mono}(t, i)$ use no error notion at all. The family $\mathrm{pred}(t, i)$ uses the band only at the greedy state S^t, so it stays valid with η replaced by η^{tr}, and, read directly as a ratio of true gains at greedy states, with η replaced by η^{sel}. The family $\mathrm{cons}(t, i)$ invokes Lemma 9, whose proof uses the band at the state $S^t \cup \{o_i\}$, which is off the greedy trajectory; so* cons *is not covered by either restricted measure without a further assumption. This is the reason the L_K bound, whose proof uses only* sum *and* pred*, is stated for η^{sel} in Proposition 6, while the exact value of Theorem 10 is stated for the global η.*

Remark 34 (which hypothesis each family consumes)**.** $\mathrm{sum}(t)$ *uses submodularity and monotonicity of f and no property of $\tilde{f}$.* $\mathrm{pred}(t, i)$ *uses the greedy choice and both bands, and no property of f beyond monotonicity.* $\mathrm{mono}(t, i)$ *uses submodularity, and monotonicity in case (c).* $\mathrm{cons}(t, i)$ *uses the greedy choice and both bands through Lemma 9, and monotonicity in cases (a) and (c). No family uses submodularity of $\tilde{f}$, which is consistent with the predictor being an arbitrary set function.*

The slack identities of pred and cons case (b) are machine-checked by `results/H3_j2_recheck.py` and `results/J2_core_oracles.py` (symbolic, plus 1,536 full-lattice slack-minimization targets); the bookkeeping around them is not oracle-checked. The linear program the subsection justifies is compared against the full-lattice program by `results/N3_K5_lattice.py`.

B.14 Instances used in the experiments

The worst-case instances run in Section 5 are exactly the families of Appendices B.4 and B.6: the family equation 1 at $\hat{a} = 2$, and the family equation 11 for $K \in \{3, 5, 8\}$ with one η per segment j. The experiment pipeline runs them through the same lazy-greedy code as the real objectives, with the element order implementing the adversarial tie direction (B before O, and C then P before O), and reproduces the values $L_K(\hat{a})$ and $V_j(\eta)$ of the two previous subsections to 10^{-10}; on every V_j instance it measures $\eta^{\mathrm{sel}} = \eta$, and on every member of equation 1 it measures $\eta^{\mathrm{sel}} = \eta^{\mathrm{tr}} = \hat{a} < \eta$.

The computations of this subsection are machine-checked by `results/E4_worst_instances.py` and `results/F2_etasel_tight.py`.

C Experiment details and additional figures

C.1 Protocol

Every run is a single trajectory of predictive greedy on $\tilde{f}$ of length equal to the largest budget of its family, from which all shorter budgets are read off as prefixes. Alongside it the same code runs greedy on f to provide the ratio denominator. Both objectives are wrapped in a cache keyed by the set, so a repeated query is free and the running times quoted in the released notes count distinct sets.

At every state of the $\tilde{f}$ trajectory the true best gain $\max_{e \notin S^t} d_e(S^t)$ is computed as well, which is what makes η^{sel} measurable; where f is not submodular this maximum is taken by an exact scan rather than by a lazy heap, since lazy evaluation is only valid under submodularity. Steps whose chosen true gain is nonpositive are counted separately (last column of Table 1); under Definition 3 a harmful such step makes the stepwise η^{sel} infinite, which the in-model family reports as certificate 0, while the out-of-model families print the finite-step diagnostic instead.

The trajectory error η^{tr} is measured on the same runs, but only over candidate pairs whose true and predicted gains both exceed one quantization unit of the objective, which is one over the size of the held-out split for feature selection and one covered node for influence maximization. Pairs below that threshold are exactly the ones that send a multiplicative ratio to infinity for reasons that have nothing to do with the selection, so the figures reported for η^{tr} are lower estimates of it, and the bound $L_K(\eta^{\text{tr}})$ computed from them is an upper estimate of the bound the untrimmed measure would give. Both directions favour η^{tr}, which is why the comparison with η^{sel} in Section 5 is stated in the direction it is.

For feature selection the predictor is the 5-fold cross-validation accuracy on the training split only; the held-out split that defines f is never touched by $\tilde{f}$. For influence maximization each observed graph is drawn once per (network, p, seed) and reused along the whole trajectory, so $\tilde{f}$ is a fixed function and not a fresh sample per query. For summarization the reference summaries are held out of every surrogate.

C.2 How good is the OPT proxy?

Ratios in Table 1 divide by greedy on the true objective rather than by the optimum, which makes them upper estimates. On the smallest feature-selection dataset the optimum was enumerated exactly for budgets up to $K = 5$ over 10 splits. At that budget the median of $f(S^f)/f(O^*)$ is 0.982, so using the proxy inflates a reported ratio by 1.8 % in the median; the median of $f(S^{\tilde{f}})/f(O^*)$, that is the quantity the theory bounds, is 0.942. Exhaustive enumeration was not attempted at larger budgets or on the larger datasets, so no ratio against a true optimum is claimed anywhere else. On the largest dataset a conservative under-estimate $\hat{O}$ of the optimum was computed as the best true value over 2200 candidate subsets per split (the surrogate's top-ranked subsets together with uniformly random ones), over 10 splits: the pool improved on greedy on f in 3 splits, never by more than 0.33 %. Since $f(\hat{O}) \leq f(O^*)$, this bounds how much better the sampled pool is, not the gap to the true optimum, and the table keeps greedy on f as the denominator.

C.3 Baseline comparison

The baseline panel uses 10 splits rather than the 30 of the main sweep, to bound running time. On the largest feature-selection dataset the median held-out accuracy of greedy on $\tilde{f}$ is at least that of the best of 4 classical selectors (a mutual-information ranking in the max-relevance tradition (Peng et al., 2005), a univariate-statistic ranking, recursive feature elimination, and the feature importances of extremely randomized trees (Geurts et al., 2006)) at 7 of the 7 budgets, all of them scored by the same f so that the comparison is between selected subsets and not between downstream models. On the smallest dataset the outcome is mixed: greedy on $\tilde{f}$ leads at some budgets and trails at others, in every case by an amount on the order of a single accuracy quantization unit of that dataset, so the fair statement there is a draw rather than an improvement.

C.4 Additional figures

C.5 Constructed instances

The 19 constructed instances of Section 5 consist of 16 instances of the V_j family, one per segment of Theorem 10 at $K \in \{3, 5, 8\}$, and 3 instances of the per-K family of Theorem 8. On the V_j instances the measured selection error reproduces the design error to 5.3×10^{-15}; on the family of Theorem 8 the measured η^{sel} reproduces the parameter $\hat{a}$ of that family to 1.8×10^{-15}, which is the equality $\eta^{\text{sel}} = \hat{a}$ asserted there and not the global error of those instances, which is larger. The representative row of Table 1 is the $j = 2$ instance at $K = 5$ and error 3.5, whose realized ratio

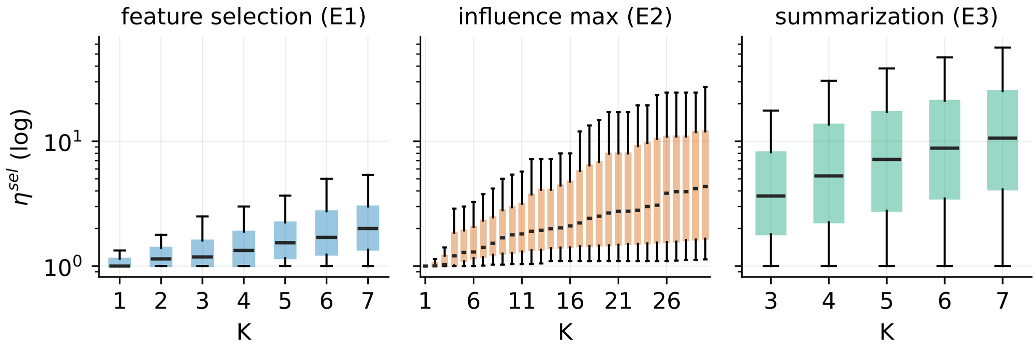

Figure 3: Distribution of the finite-step selection diagnostic by budget K for each surrogate family (boxes: quartiles over runs; log scale; outliers suppressed; the influence-maximization runs whose stepwise η^{sel} is infinite enter the text counts, not these boxes). The learned surrogates stay in the single digits over the whole budget range, while the heuristic surrogates do not.

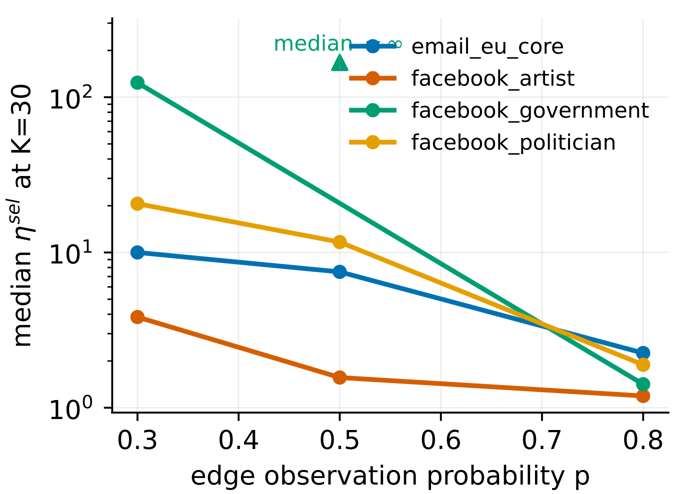

Figure 4: Median stepwise η^{sel} at $K = 30$ against the edge observation probability p, one line per network (log scale). The error rises monotonically as the graph is observed less on 3 of the 4 networks; on the remaining network one cell has an infinite median (marker at the top: half its runs contain a harmful zero step and certify nothing). Over the same range the realized ratio falls only from no worse than 0.988 to between 0.882 and 0.963, while the certified bound $L_{30}(\eta^{\text{sel}})$ falls by up to a factor 64 (and to 0 on the infinite cell). Source: results/K3_split_figs.py.

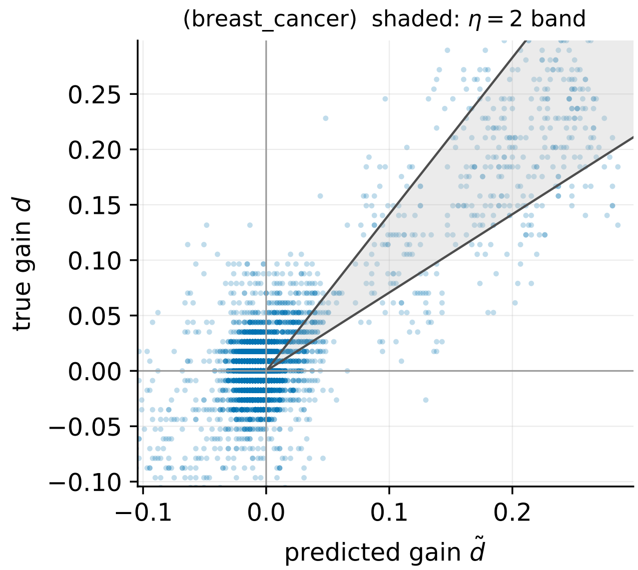

Figure 5: True gain d against predicted gain $\tilde{d}$ over the candidate evaluations of one feature-selection dataset, with a multiplicative error band shaded. Sign disagreements concentrate near zero gain, where the accuracy quantization of the objective dominates, and the large gains are ordered correctly. This is why the guarantee is read at η^{sel}, which charges only the step that was taken, rather than at the band error of every candidate pair.

is 0.278. No constructed instance produced a sign disagreement between predicted and true gains (largest measured share 0 %).